

# MATERNAL RISK PREDICTION

## Machine Learning Based Classification Analysis

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## 1. Introduction

Maternal health is one of the most critical areas of modern healthcare systems because complications during pregnancy can directly affect both maternal and fetal survival. Early identification of pregnancy-related risks allows healthcare professionals to provide preventive treatment, continuous monitoring, and timely medical intervention. However, analyzing large amounts of medical information manually can be difficult, time-consuming, and prone to human error. For this reason, machine learning techniques have become increasingly important in healthcare applications and clinical decision-support systems.

Machine learning algorithms can analyze complex medical datasets, detect hidden patterns, and generate predictive models capable of assisting healthcare professionals in risk assessment processes. In maternal healthcare, factors such as blood sugar level, blood pressure, body mass index (BMI), heart rate, and preexisting diseases can significantly influence pregnancy outcomes. By learning relationships between these medical indicators and maternal risk levels, machine learning models can help improve prediction accuracy and support early diagnosis.

The primary objective of this project is to develop and evaluate multiple machine learning classification models for maternal risk prediction using a structured medical dataset. The dataset contains various physiological and medical measurements collected from pregnant individuals, along with a target variable representing maternal risk level.

In this study, a complete machine learning pipeline was developed, including:

- exploratory data analysis (EDA),
- missing value handling,
- feature scaling,
- target variable encoding,
- model training,
- model evaluation,
- cross-validation analysis,
- and feature importance interpretation.

Several supervised machine learning algorithms were implemented and compared, including:

- Logistic Regression,
- K-Nearest Neighbors (KNN),
- Support Vector Machine (SVM),
- Decision Tree,
- and Random Forest.

The performance of these models was evaluated using multiple classification metrics such as accuracy, ROC-AUC score, precision, recall, F1-score, confusion matrices, and cross-validation scores. The goal was not only to identify the most accurate predictive model, but also to analyze which medical features contribute most strongly to maternal risk prediction.

In supervised machine learning classification, the prediction process can generally be represented as:

$$\hat{y} = f(X)$$

where:

- $X$  represents the input medical features,
- $f$  represents the learned machine learning model,
- and  $\hat{y}$  represents the predicted maternal risk level.

The findings of this project demonstrate that machine learning approaches can achieve highly successful classification performance for maternal risk prediction tasks. The results also highlight the importance of specific medical indicators such as preexisting diabetes, blood sugar level (BS), and BMI in determining pregnancy-related risks.

## 2. Dataset Description

The dataset used in this project contains maternal health-related medical measurements collected for pregnancy risk assessment. The primary purpose of the dataset is to classify pregnant individuals into different maternal risk categories based on physiological and medical indicators.

The dataset consists of:

- 1205 observations (rows)
- 12 attributes (columns)

Among these attributes:

- 11 columns represent input medical features,
- and 1 column represents the target variable (Risk Level).

The dataset includes both continuous numerical variables and binary medical-condition indicators. These variables provide important information regarding maternal physiological condition, chronic diseases, and pregnancy-related complications.

The target variable in the dataset is:

- **Risk Level**

This variable indicates whether a patient belongs to:

- Low Risk
- High Risk

For machine learning implementation, the target variable was transformed into numerical form:

| <i>Original Label</i> | <i>Encoded Value</i> |
|-----------------------|----------------------|
| <i>Low</i>            | 0                    |
| <i>High</i>           | 1                    |

The dataset features and their medical meanings are summarized below.

## 2.1 Dataset Features

| <i>Feature Name</i>           | <i>Description</i>                         |
|-------------------------------|--------------------------------------------|
| <i>Age</i>                    | Age of the pregnant individual             |
| <i>Systolic BP</i>            | Systolic blood pressure value              |
| <i>Diastolic</i>              | Diastolic blood pressure value             |
| <i>BS</i>                     | Blood sugar level                          |
| <i>Body Temp</i>              | Body temperature measurement               |
| <i>BMI</i>                    | Body Mass Index                            |
| <i>Previous Complications</i> | Previous pregnancy complications history   |
| <i>Preexisting Diabetes</i>   | Presence of diabetes before pregnancy      |
| <i>Gestational Diabetes</i>   | Diabetes developed during pregnancy        |
| <i>Mental Health</i>          | Mental health condition indicator          |
| <i>Heart Rate</i>             | Heart rate measurement                     |
| <i>Risk Level</i>             | Target variable representing maternal risk |

## 2.2 Dataset Characteristics

The dataset contains several medically significant indicators associated with maternal risk prediction. In particular:

- diabetes-related variables,
- cardiovascular measurements,
- metabolic indicators,
- and physiological measurements

are expected to contribute strongly to classification performance.

The dataset includes:

- continuous numerical variables,
- binary categorical indicators,
- and medically correlated features.

Several features also contain outlier values, especially in:

- Blood Sugar (BS),
- BMI,
- and Blood Pressure measurements.

However, these outliers were preserved during preprocessing because they may represent clinically meaningful high-risk medical conditions rather than data collection errors.

The machine learning classification process attempts to learn the relationship between medical measurements and maternal risk level. Mathematically, the dataset can be represented as:

$$D = \{(x_i, y_i)\}_{i=1}^n$$

where:

- $x_i$  represents the medical feature vector of an individual patient,
- $y_i$  represents the corresponding maternal risk label,
- and  $n$  represents the total number of observations in the dataset.

### 2.3 Dataset Importance

Maternal health datasets are highly important for healthcare analytics because early detection of pregnancy-related complications can significantly reduce maternal and fetal mortality rates.

The availability of multiple physiological and medical variables makes this dataset suitable for:

- supervised machine learning,
- classification analysis,
- predictive healthcare systems,
- and medical decision-support applications.

This dataset provides a strong foundation for evaluating how different machine learning algorithms perform on healthcare-oriented classification problems.

## 3. Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) is a critical stage in machine learning projects because it helps reveal hidden patterns, identify anomalies, understand feature distributions, and evaluate relationships between variables before model training.

In this project, EDA was performed to:

- analyze the overall structure of the dataset,
- identify missing values,
- investigate feature distributions,
- detect outliers,
- examine correlations between variables,
- and evaluate how medical indicators differ across maternal risk groups.

The analysis provides important insights into the medical relevance of the dataset and supports the preprocessing and model selection stages of the machine learning pipeline.

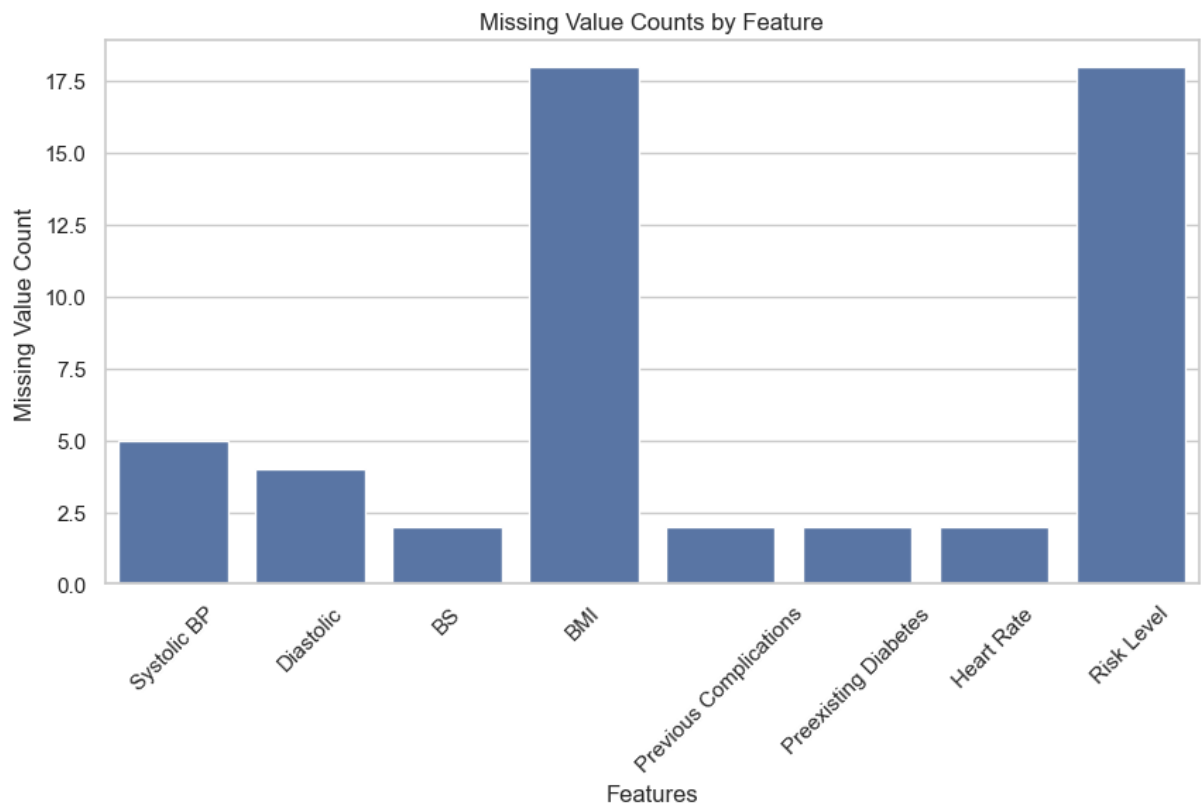
### 3.1 Missing Value Analysis

The first step of EDA involved identifying missing values within the dataset. Missing data can negatively affect machine learning performance and may lead to biased predictions if not handled correctly.

The analysis showed that several features contained missing observations, particularly:

- BMI,
- Risk Level,
- and Systolic Blood Pressure.

The number of missing values was relatively small compared to the total dataset size; therefore, imputation techniques were applied instead of removing observations.



Missing value counts for each feature in the maternal risk dataset.

The visualization indicates that:

- BMI and Risk Level contain the highest number of missing values,
- while several medical measurements contain only a small number of incomplete observations.

To preserve dataset size and maintain statistical consistency:

- numerical missing values were replaced using median imputation,

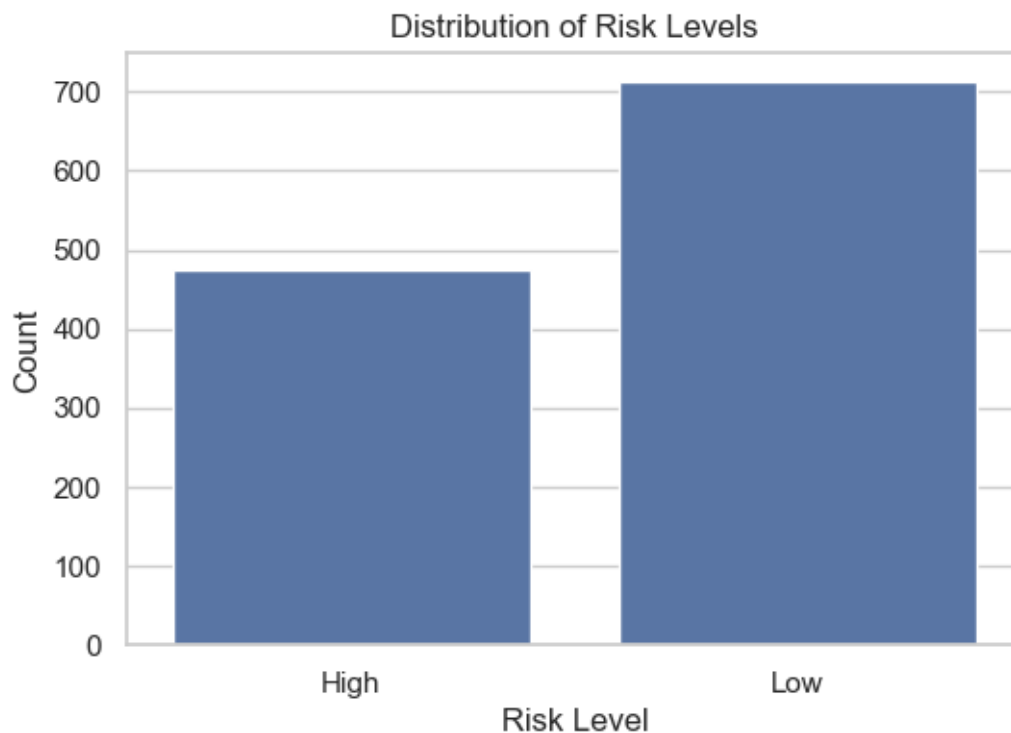
- and categorical target values were handled using mode imputation.

### 3.2 Target Variable Distribution

Understanding the distribution of target classes is important in classification problems because severe class imbalance may negatively affect model performance.

The maternal risk dataset contains two target classes:

- High Risk
- Low Risk



Distribution of maternal risk categories in the dataset.

The visualization demonstrates that:

- the Low Risk category contains a larger number of observations,
- while the High Risk category contains fewer samples.

Although the dataset is not perfectly balanced, the class distribution is still sufficiently suitable for supervised classification modeling.

The probability distribution of a binary classification problem can generally be represented as:

$$P(Y = 0) + P(Y = 1) = 1$$

where:

- $Y = 0$  represents the Low Risk class,
- and  $Y = 1$  represents the High Risk class.

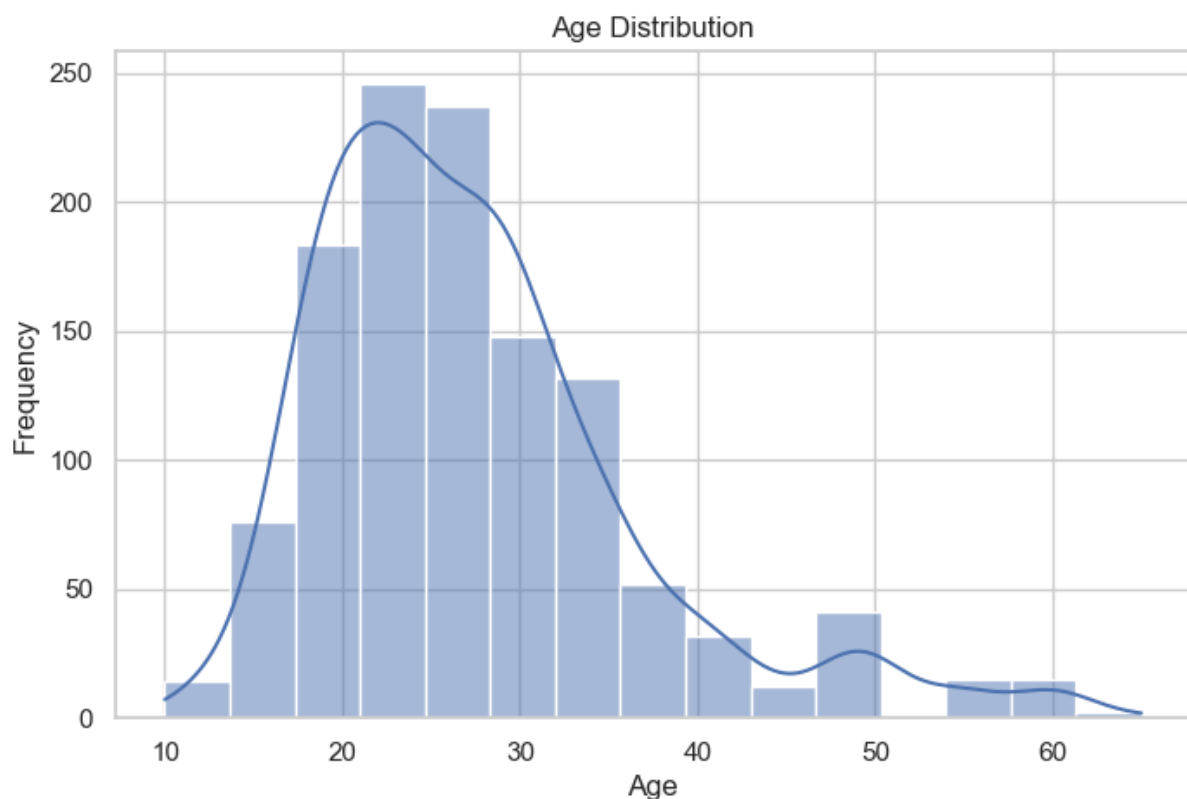
### 3.3 Distribution Analysis

Distribution analysis helps evaluate whether variables follow normal or skewed patterns and allows identification of medically unusual observations.

The distributions of several important medical indicators were analyzed, including:

- Age,
- BMI,
- Systolic Blood Pressure,
- and Diastolic Blood Pressure.

#### Age Distribution



Distribution of patient age values.

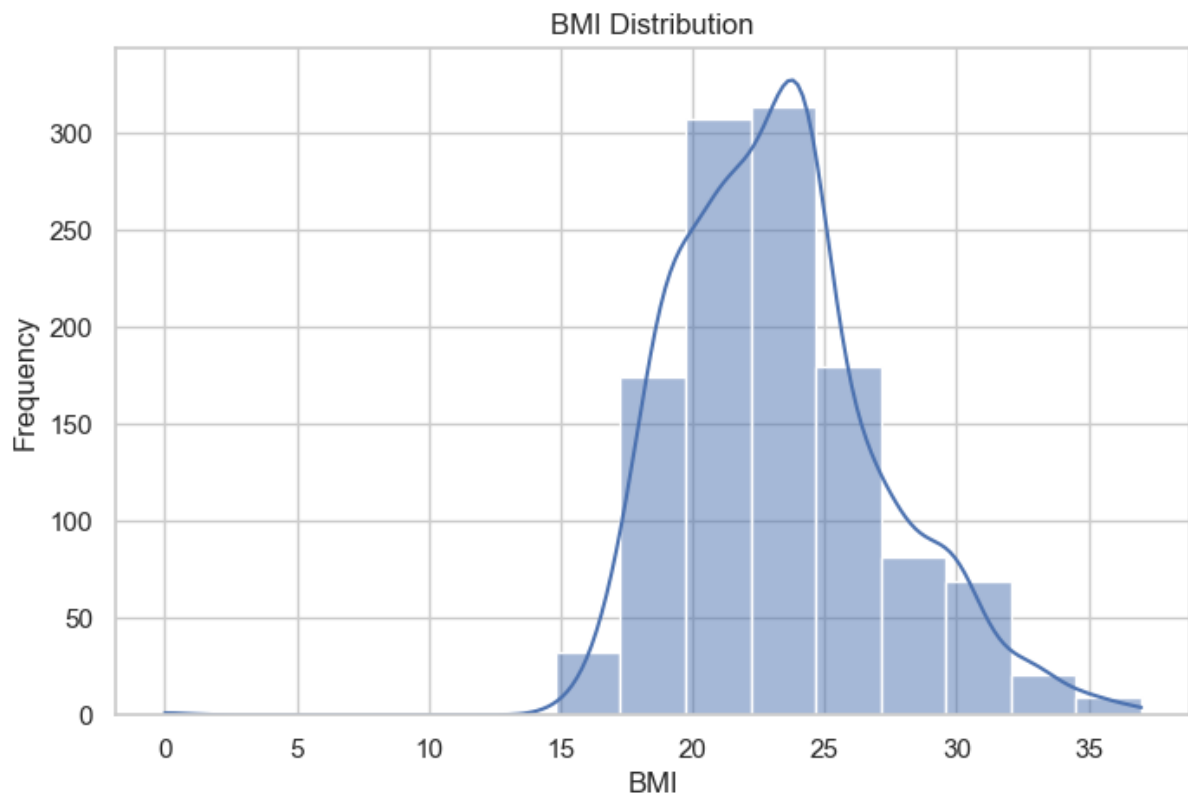
The age distribution shows that:

- most patients are concentrated between approximately 20 and 35 years old,
- while older patients appear less frequently.



The distribution is slightly right-skewed due to several higher-age observations.

### BMI Distribution

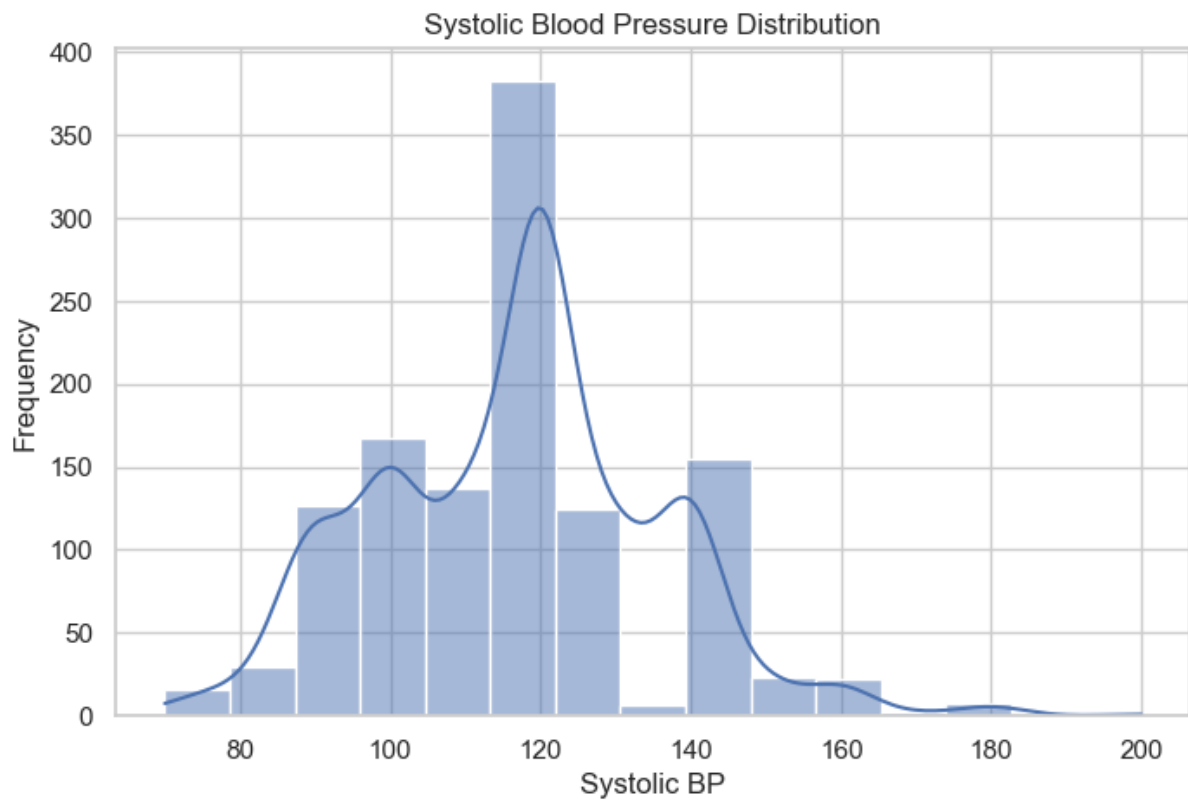


Distribution of Body Mass Index (BMI) values.

The BMI distribution appears approximately normal around the central region, although:

- several high BMI observations exist,
- suggesting potential obesity-related maternal health risks.

## Systolic Blood Pressure Distribution



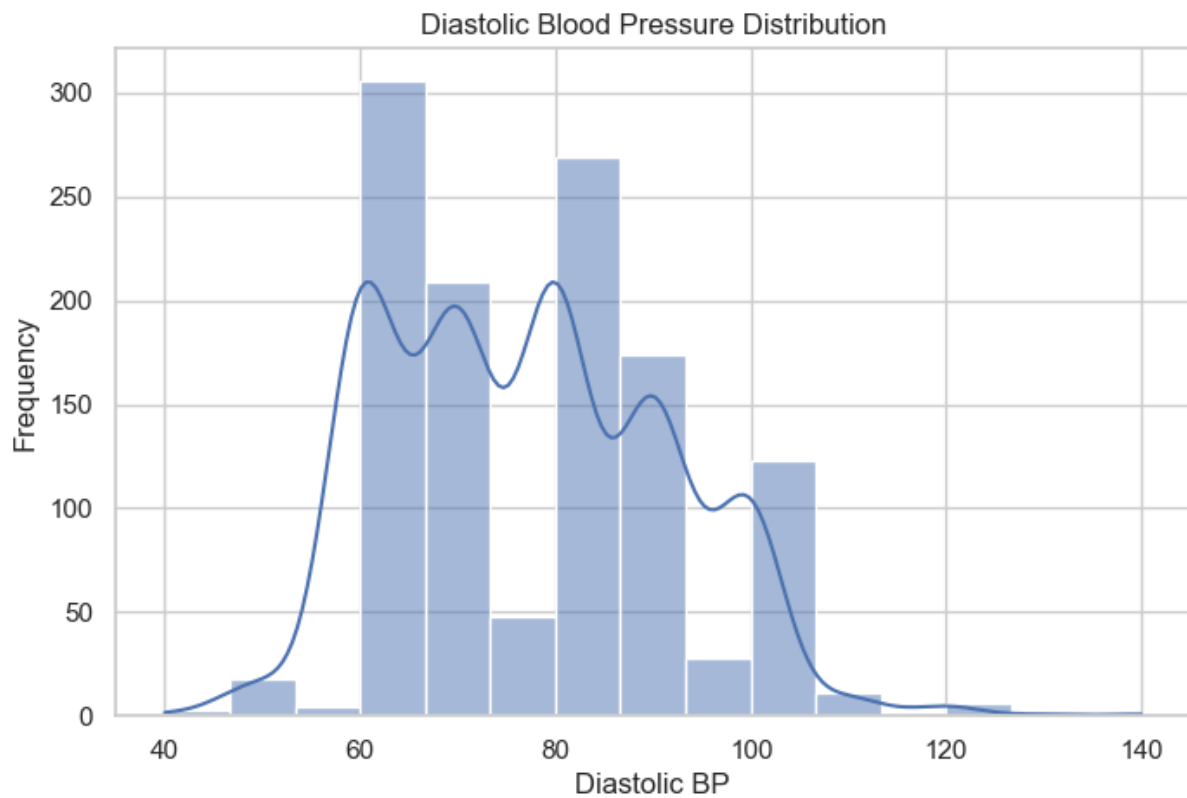
Distribution of systolic blood pressure values.

The majority of systolic blood pressure values are concentrated around:

- 100–140 mmHg.

However, several high-pressure observations indicate potential hypertension-related risk conditions.

## Diastolic Blood Pressure Distribution



Distribution of diastolic blood pressure values.

The diastolic blood pressure values are centered around:

- 60–90 mmHg,

with several higher outlier observations.

### 3.4 Outlier Analysis Using Boxplots

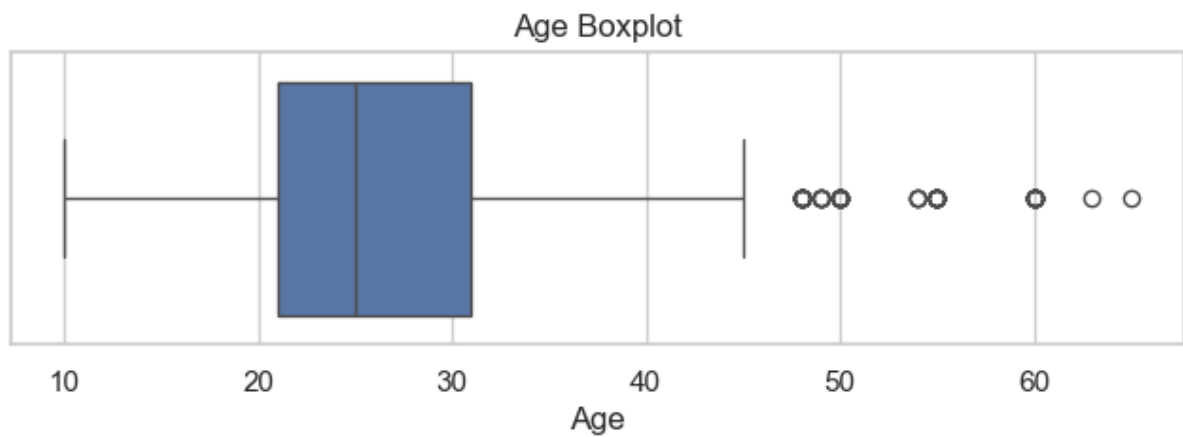
Boxplots were used to identify extreme observations and evaluate data variability.

In medical datasets, outliers may represent:

- measurement errors,
- or clinically significant high-risk patients.

Therefore, outliers were not automatically removed during preprocessing.

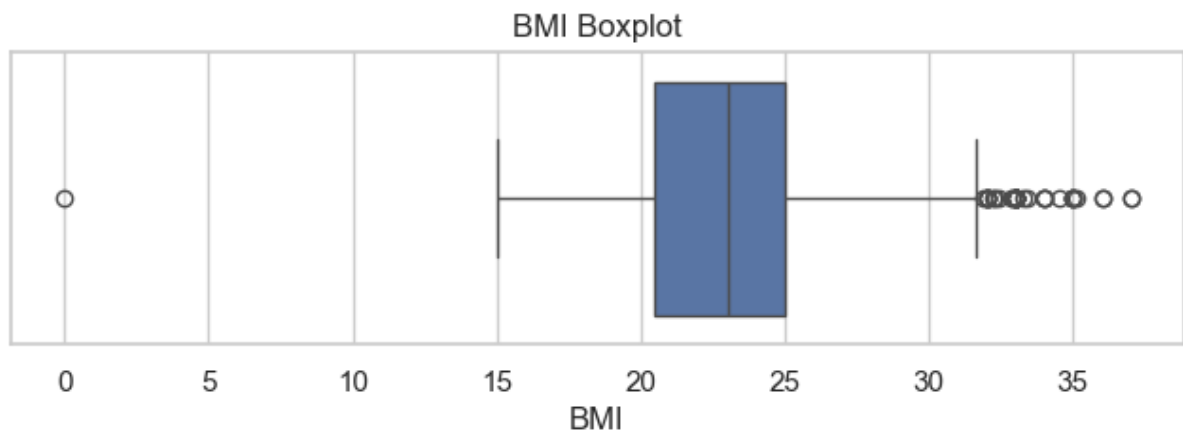
### Age Boxplot



Age boxplot showing the presence of high-age outliers.

Several higher-age observations appear beyond the upper whisker, indicating possible elevated pregnancy risk among older patients.

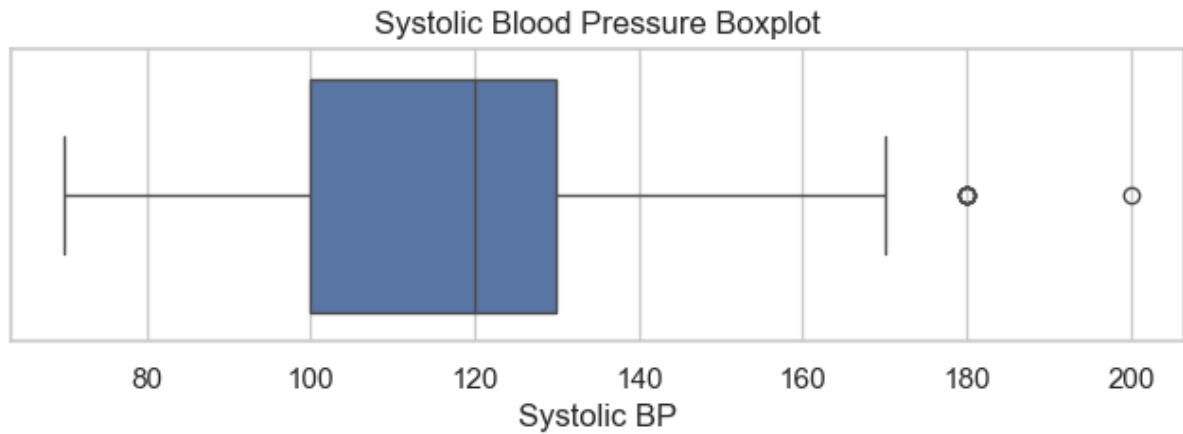
### BMI Boxplot



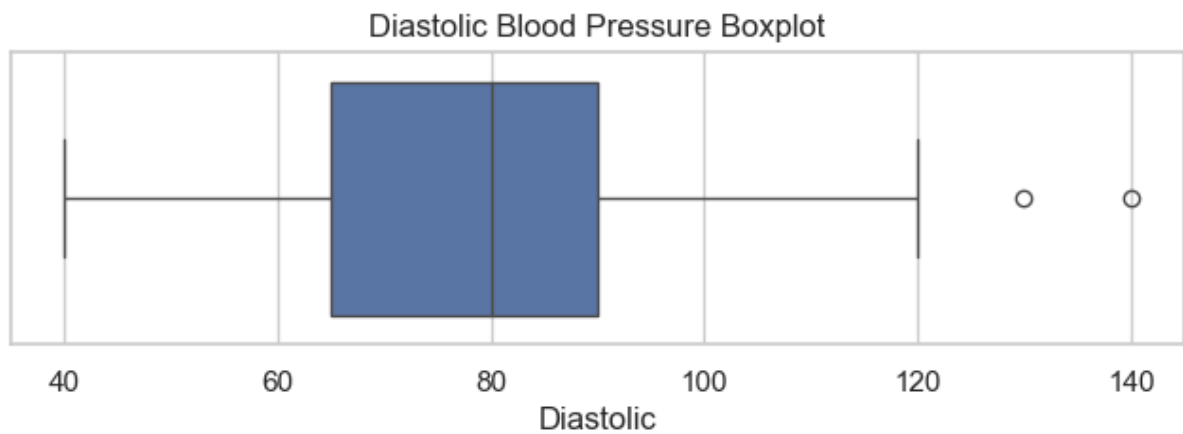
BMI boxplot illustrating high-BMI outliers.

The visualization reveals several extreme BMI values associated with obesity-related maternal complications.

## Blood Pressure Boxplots



Systolic blood pressure outlier analysis.



Diastolic blood pressure outlier analysis.

The boxplots demonstrate the existence of several elevated blood pressure observations that may correspond to hypertensive pregnancy conditions.

## 3.5 Correlation Analysis

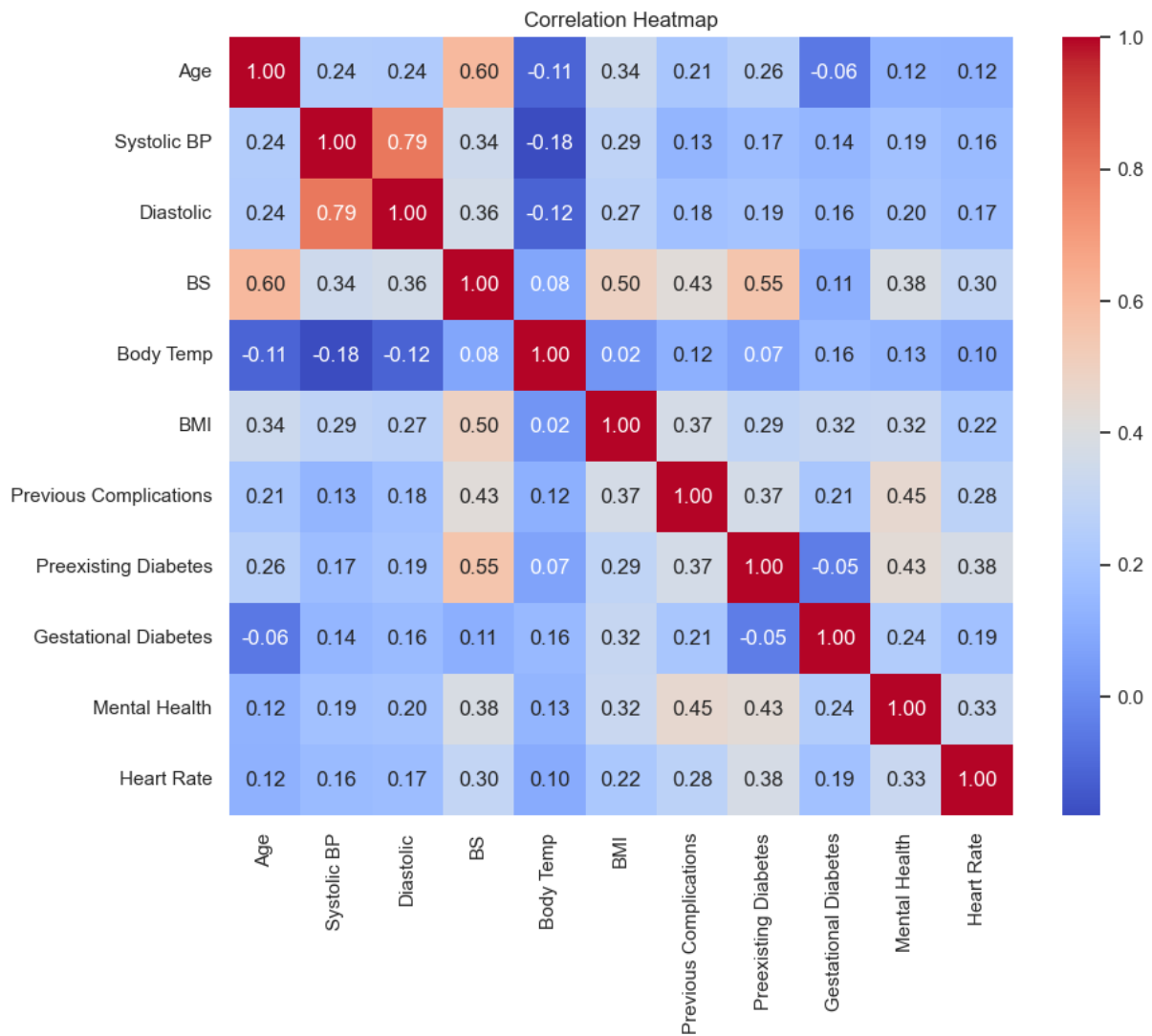
Correlation analysis was performed to examine relationships between medical variables.

The Pearson correlation coefficient measures linear relationships between variables and can be mathematically expressed as:

$$r = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum(x_i - \bar{x})^2 \sum(y_i - \bar{y})^2}}$$

where:

- $r$  represents the correlation coefficient,
- and values closer to 1 indicate stronger positive relationships.



Correlation heatmap between medical features.

Important observations include:

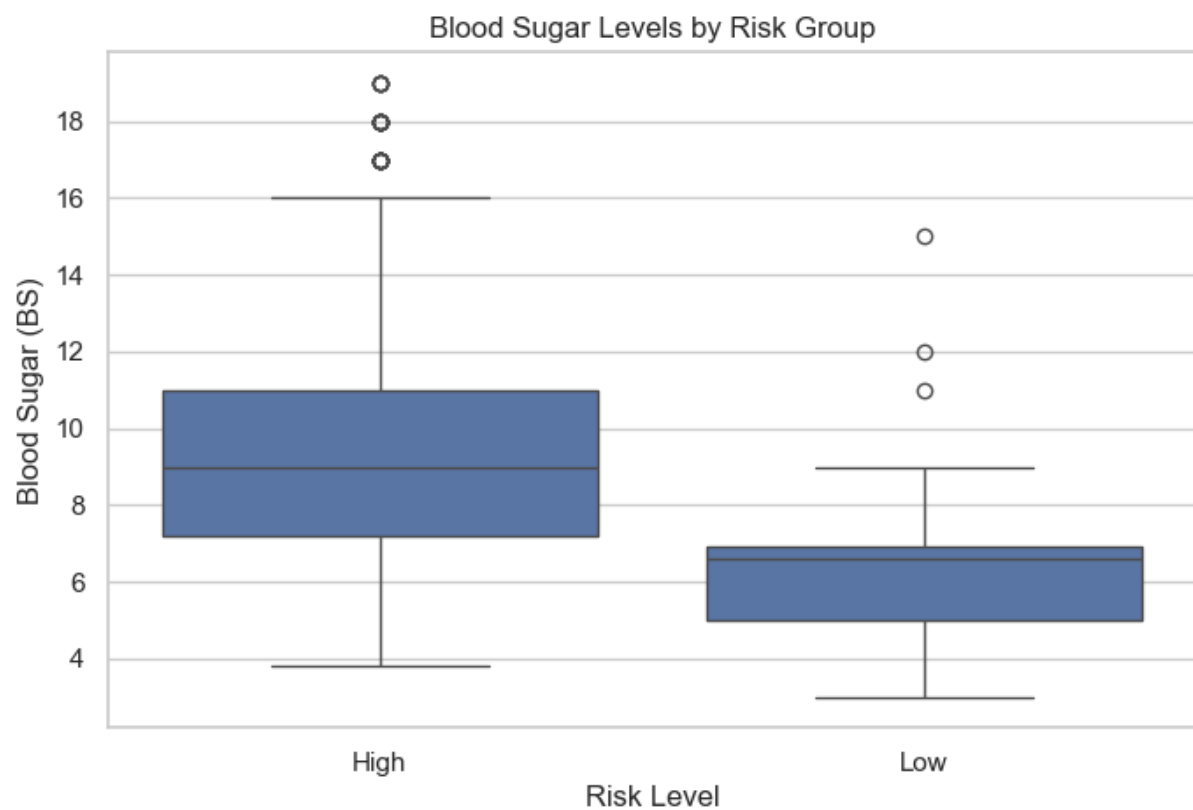
- strong correlation between systolic and diastolic blood pressure,
- moderate correlation between BS and preexisting diabetes,
- and several medically meaningful relationships among cardiovascular indicators.

The correlation structure confirms that the dataset contains informative medical patterns suitable for machine learning classification.

### 3.6 Risk-Level-Based Comparative Analysis

Medical indicators were also analyzed separately according to maternal risk groups.

## Blood Sugar by Risk Group



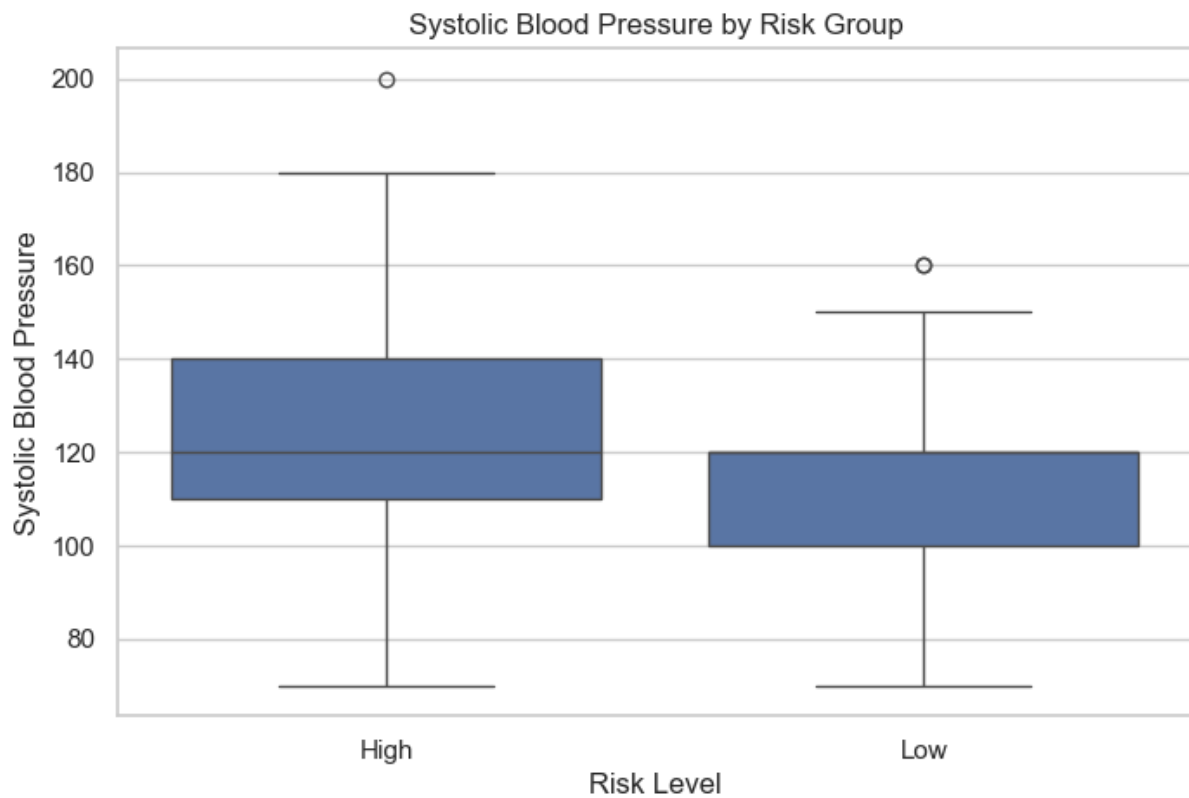
Blood sugar (BS) distribution across maternal risk groups.

The High Risk group demonstrates:

- noticeably higher blood sugar levels,
- larger variability,
- and more extreme observations.

This suggests that blood sugar level is a highly influential predictor of maternal risk.

### Systolic Blood Pressure by Risk Group



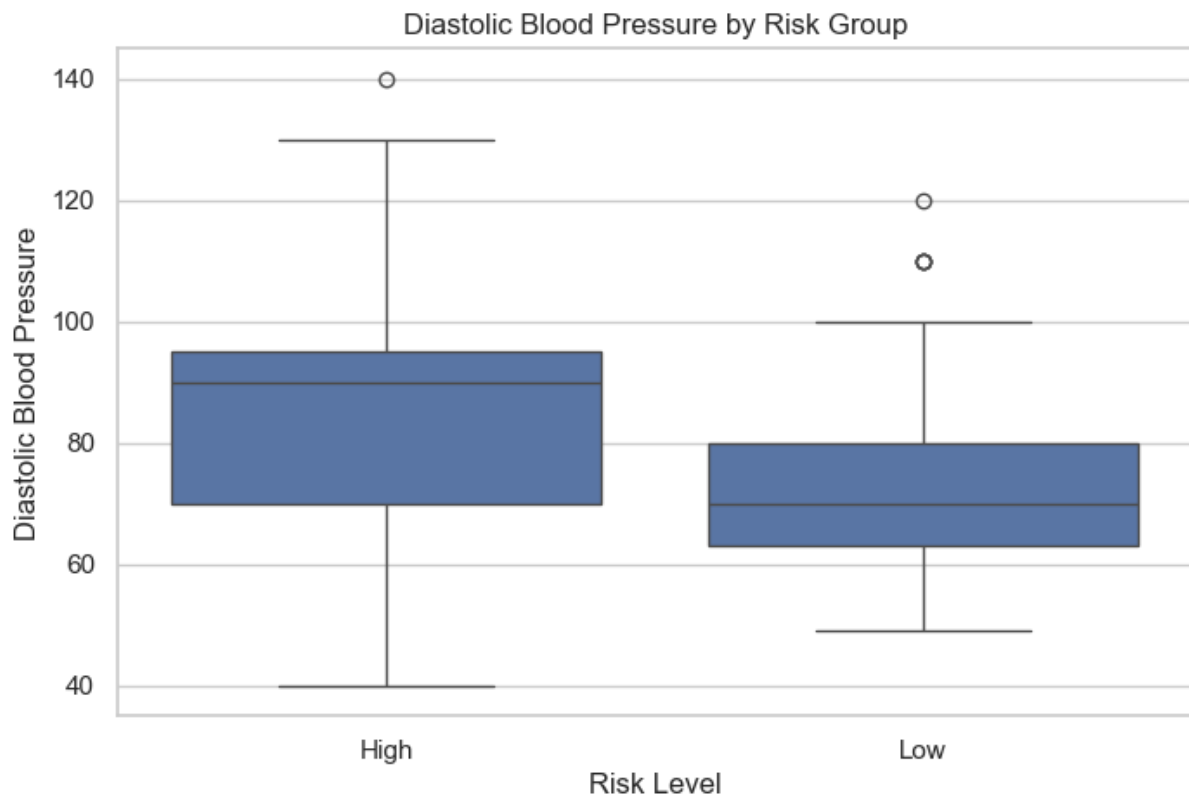
Systolic blood pressure comparison across maternal risk groups.

Patients in the High Risk category generally exhibit:

- higher systolic blood pressure values,
- and greater variability compared to the Low Risk group.



### Diastolic Blood Pressure by Risk Group



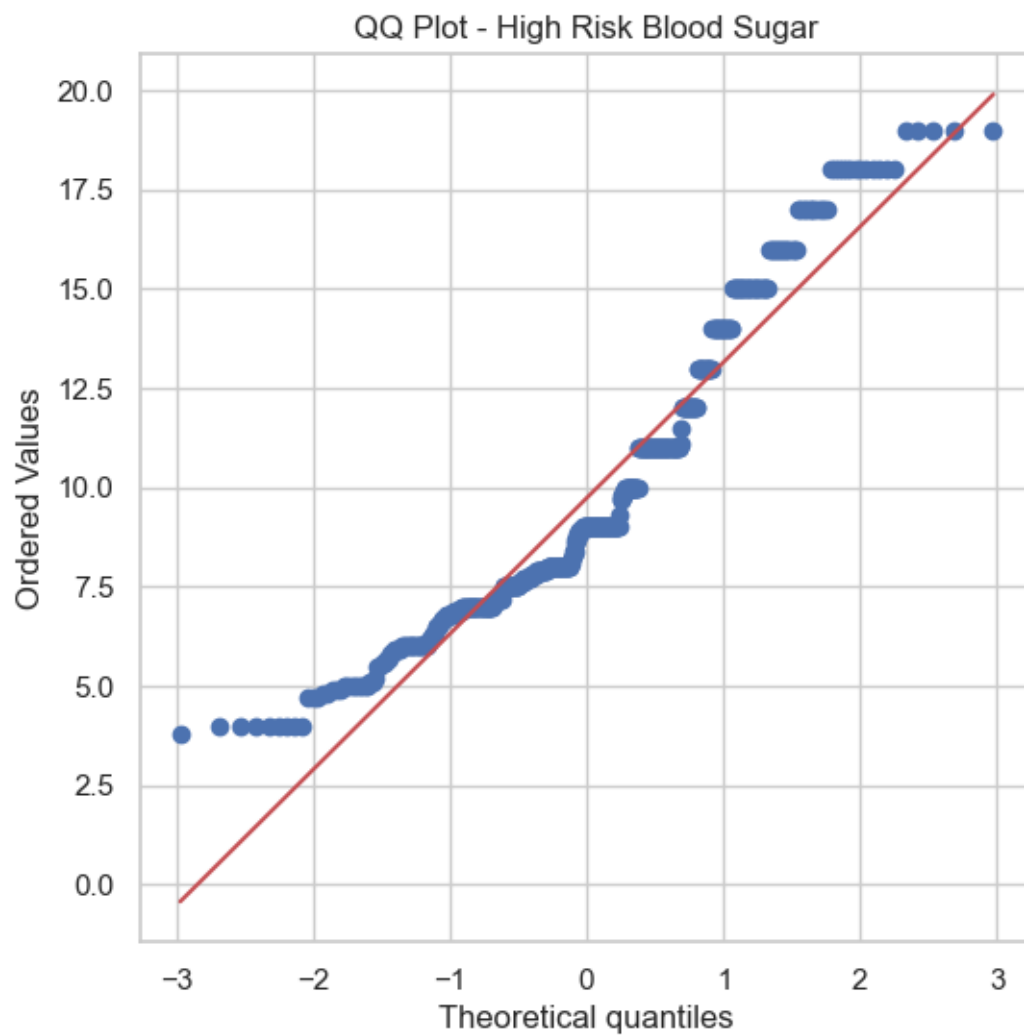
Diastolic blood pressure comparison across maternal risk groups.

The High Risk group also demonstrates elevated diastolic blood pressure values relative to the Low Risk group.

### 3.7 QQ Plot Analysis

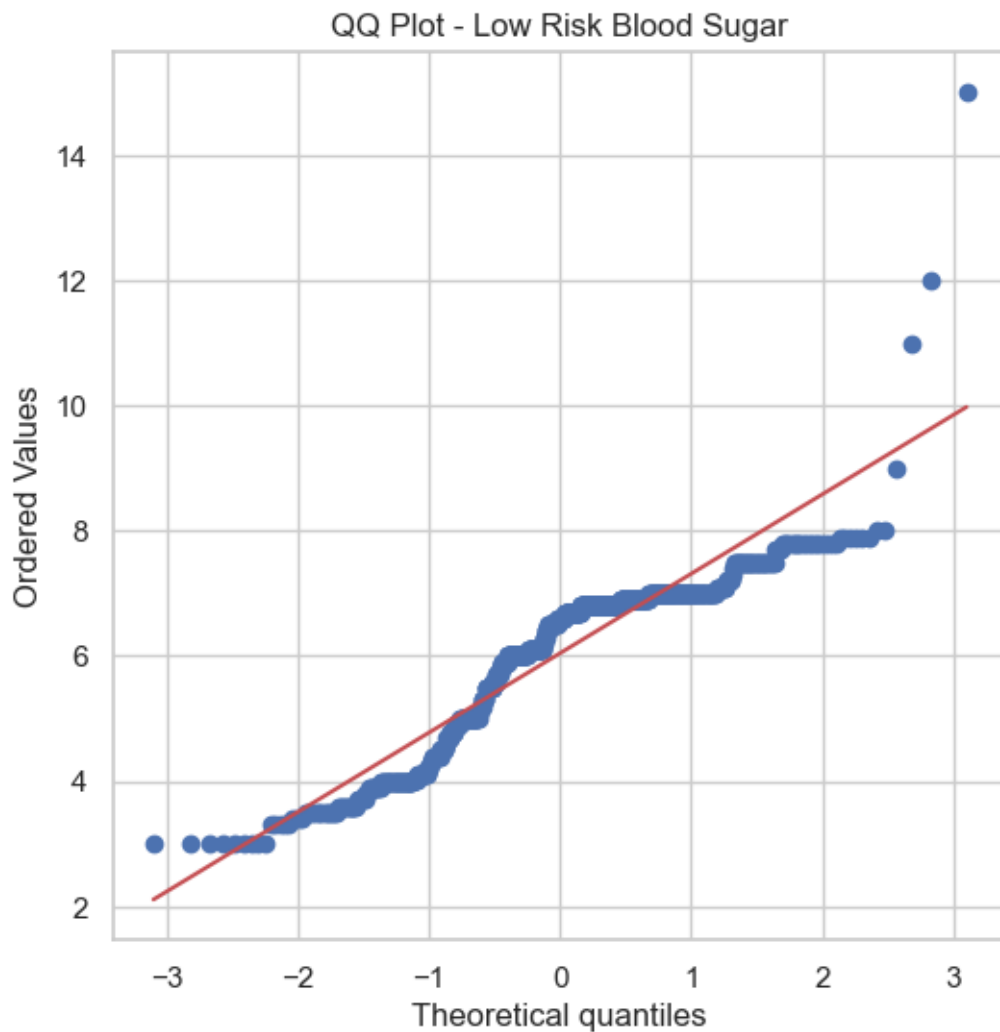
QQ plots were used to evaluate whether blood sugar values approximately follow a normal distribution.

### High Risk Blood Sugar QQ Plot



QQ plot for blood sugar levels in the High Risk group.

### Low Risk Blood Sugar QQ Plot



QQ plot for blood sugar levels in the Low Risk group.

The QQ plots indicate that:

- blood sugar values deviate from perfect normality,
- especially in the upper quantiles,
- suggesting the presence of skewness and medically extreme observations.

### EDA Summary

The exploratory data analysis revealed several important findings:

- blood sugar and diabetes-related variables strongly influence maternal risk,
- high-risk patients generally exhibit higher blood pressure measurements,
- several clinically meaningful outliers exist in the dataset,
- and the dataset contains sufficiently informative relationships for machine learning classification.

The findings obtained during EDA directly guided the preprocessing strategy and machine learning modeling stages of the project.

## 4. Data Preprocessing

Data preprocessing is one of the most critical stages in machine learning because the quality of input data directly affects model performance and prediction reliability.

The maternal health dataset contained:

- missing values,
- categorical target labels,
- features with different numerical scales,
- and variables requiring transformation before model training.

Therefore, several preprocessing techniques were applied to prepare the dataset for classification algorithms.

The preprocessing pipeline included:

1. Missing value handling,
2. Target encoding,
3. Feature-target separation,
4. Train-test splitting,
5. and feature scaling.

These operations ensured that the dataset became:

- numerically consistent,
- machine-learning compatible,
- and statistically reliable for predictive modeling.

### 4.1 Missing Value Handling

Missing values are common in real-world medical datasets and can significantly reduce model performance if not handled properly.

Instead of removing incomplete observations, imputation techniques were applied to preserve the dataset size and maintain information consistency.

Two different imputation strategies were used depending on the data type:

- **Median Imputation** for numerical features
- **Mode Imputation** for categorical variables

## Median Imputation

For numerical medical measurements such as:

- BMI,
- Blood Pressure,
- and Blood Sugar,

missing values were replaced using the median of the corresponding feature.

The median is mathematically defined as:

$$\tilde{x} = \begin{cases} x_{(\frac{n+1}{2})}, & \text{if } n \text{ is odd} \\ x_{(\frac{n}{2})} + x_{(\frac{n}{2}+1)}, & \text{if } n \text{ is even} \end{cases}$$

Median imputation was preferred because:

- medical datasets often contain outliers,
- and the median is more robust than the mean in skewed distributions.

For example:

- extremely high blood pressure values would disproportionately affect mean imputation,
- whereas median imputation preserves central tendency more reliably.

## Mode Imputation

Categorical variables were handled using mode imputation.

The mode represents the most frequently occurring category:

$$Mode = \arg \max_x f(x)$$

where:

- $f(x)$  represents the frequency of category xxx.

The missing values in the Risk Level column were replaced using the most common class label.

This approach preserved dataset balance while minimizing information loss.

## 4.2 Target Encoding

Machine learning algorithms require numerical target values; therefore, categorical maternal risk labels were transformed into binary numerical values.

The encoding strategy was defined as:

| <i>Original Label</i> | <i>Encoded Value</i> |
|-----------------------|----------------------|
| <i>Low Risk</i>       | 0                    |
| <i>High Risk</i>      | 1                    |

This binary encoding simplified the classification problem into a supervised binary prediction task.

The transformation can be expressed mathematically as:

$$Y = \begin{cases} 0, & \text{Low Risk} \\ 1, & \text{High Risk} \end{cases}$$

where:

- $Y = 0$  represents low maternal risk,
- and  $Y = 1$  represents high maternal risk.

This encoding enables machine learning models to interpret the problem numerically while preserving class semantics.

#### 4.3 Feature and Target Separation

After preprocessing and encoding, the dataset was separated into:

- input features ( $X$ ),
- and target labels ( $y$ ).

The feature matrix contains all medical measurements used for prediction:

$$X = [x_1, x_2, x_3, \dots, x_n]$$

while the target vector represents maternal risk categories:

$$y = [0, 1, 0, 1, \dots]$$

The features included:

- Age,
- Blood Pressure,
- BMI,
- Blood Sugar,
- Heart Rate,
- Previous Complications,
- and other medical indicators.

Separating predictors and labels is essential because supervised machine learning algorithms learn patterns from:

- input variables ( $X$ ),
- to predict output classes ( $y$ ).

#### 4.4 Train-Test Split

The dataset was divided into:

- training data,
- and testing data.

The purpose of this separation is to evaluate whether the model can generalize to unseen data.

The split ratio used in this project was:

| <i><b>Dataset Portion</b></i> | <i><b>Percentage</b></i> |
|-------------------------------|--------------------------|
| <i>Training Set</i>           | 80%                      |
| <i>Testing Set</i>            | 20%                      |

The splitting operation can be represented as:

$$D = D_{train} + D_{test}$$

where:

- $D_{train}$  contains the observations used for learning,
- and  $D_{test}$  contains unseen observations used for evaluation.

A fixed random seed was applied:

$$random\_state = 42$$

Using a constant random state ensures:

- reproducibility,
- experimental consistency,
- and identical train-test partitions across executions.

After splitting:

- the training dataset contained 964 observations,
- while the testing dataset contained 241 observations.

## 4.5 Feature Scaling

The medical features in the dataset operate on different numerical ranges.

For example:

- Blood Pressure values may exceed 100,
- while BMI values typically remain between 15–35,
- and binary variables contain only 0 or 1 values.

Algorithms such as:

- KNN,
- Logistic Regression,
- and SVM

are highly sensitive to feature magnitudes.

Therefore, feature scaling was applied using **StandardScaler**.

### Standardization Formula

Standardization transforms variables into a distribution with:

- mean approximately equal to 0,
- and standard deviation approximately equal to 1.

The transformation formula is:

$$z = \frac{x - \mu}{\sigma}$$

where:

- $x$  = original feature value,
- $\mu$  = mean of the feature,
- $\sigma$  = standard deviation,
- $z$  = standardized value.

For example:

$$z = \frac{x - \mu}{\sigma} \approx 1.2$$

A standardized score of:

$$z \approx 1.2$$

indicates that the observation is:



- approximately 1.2 standard deviations above the mean.

Under normal distribution assumptions:

$$\Phi(z) \approx 88.5$$

This means the value lies above approximately 88.5% of observations.

### Why Feature Scaling Is Necessary

Feature scaling improves:

- convergence speed,
- numerical stability,
- and distance-based learning performance.

Without scaling:

- features with large numerical values dominate distance calculations,
- causing biased predictions.

For example, Euclidean distance in KNN is calculated as:

$$d(x, y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$$

If variables are not standardized:

- blood pressure values could dominate BMI or binary medical indicators,
- reducing model fairness and prediction quality.

Feature scaling therefore ensures:

- balanced feature contribution,
- stable optimization,
- and improved classification performance across all machine learning algorithms used in this study.

### Preprocessing Summary

The preprocessing stage successfully transformed the raw maternal health dataset into a machine-learning-ready format.

The applied operations included:

- handling missing values,

- encoding categorical labels,
- separating predictors and targets,
- splitting training and testing datasets,
- and standardizing feature scales.

These preprocessing steps improved:

- data consistency,
- statistical reliability,
- and predictive model performance.

The processed dataset was then used for training and evaluating multiple machine learning classification algorithms.

## 5. Machine Learning Models

After preprocessing the maternal health dataset, multiple supervised machine learning algorithms were implemented and evaluated for maternal risk classification.

The objective of this stage was to:

- compare different classification approaches,
- evaluate prediction performance,
- and identify the most reliable model for maternal risk prediction.

Five classification algorithms were selected:

1. Logistic Regression
2. K-Nearest Neighbors (KNN)
3. Support Vector Machine (SVM)
4. Decision Tree
5. Random Forest

These algorithms were chosen because they represent different learning paradigms:

- linear classification,
- distance-based learning,
- margin-based optimization,
- tree-based learning,
- and ensemble learning.

Each model was trained using the preprocessed dataset and evaluated using:

- Accuracy,
- ROC-AUC,
- Cross Validation,
- Confusion Matrix,

- and Classification Report metrics.

## 5.1 Logistic Regression

Logistic Regression is one of the most widely used binary classification algorithms in machine learning.

Although the name contains “regression,” the algorithm is designed for classification problems where the output belongs to one of two categories.

In this project, Logistic Regression was used to classify maternal patients into:

- Low Risk (0)
- High Risk (1)

### Logistic Regression Theory

Logistic Regression estimates the probability that a sample belongs to a specific class using the sigmoid function.

The sigmoid function transforms linear outputs into probability values between 0 and 1.

The probability equation is defined as:

$$P(y = 1) = \frac{1}{1 + e^{-z}}$$

where:

$$z = w_1x_1 + w_2x_2 + \dots + w_nx_n + b$$

- $x_i$  represents input features,
- $w_i$  represents model weights,
- and  $b$  represents the bias term.

The sigmoid function converts the linear combination into a probability score.

Classification is then performed using a threshold:

$$\hat{y} = \begin{cases} 1, & P(y = 1) \geq 0.5 \\ 0, & P(y = 1) < 0.5 \end{cases}$$

This means:

- probabilities greater than or equal to 0.5 are classified as High Risk,
- otherwise classified as Low Risk.

### Why Logistic Regression Was Used

Logistic Regression was selected because:

- it performs well on binary classification tasks,
- it is computationally efficient,
- and it provides interpretable decision boundaries.

The algorithm is especially useful in medical prediction problems because:

- probability outputs can be interpreted clinically,
- feature coefficients can indicate risk influence,
- and the model is less prone to overfitting compared to more complex methods.

Additionally, Logistic Regression works effectively after feature standardization, which was applied during preprocessing.

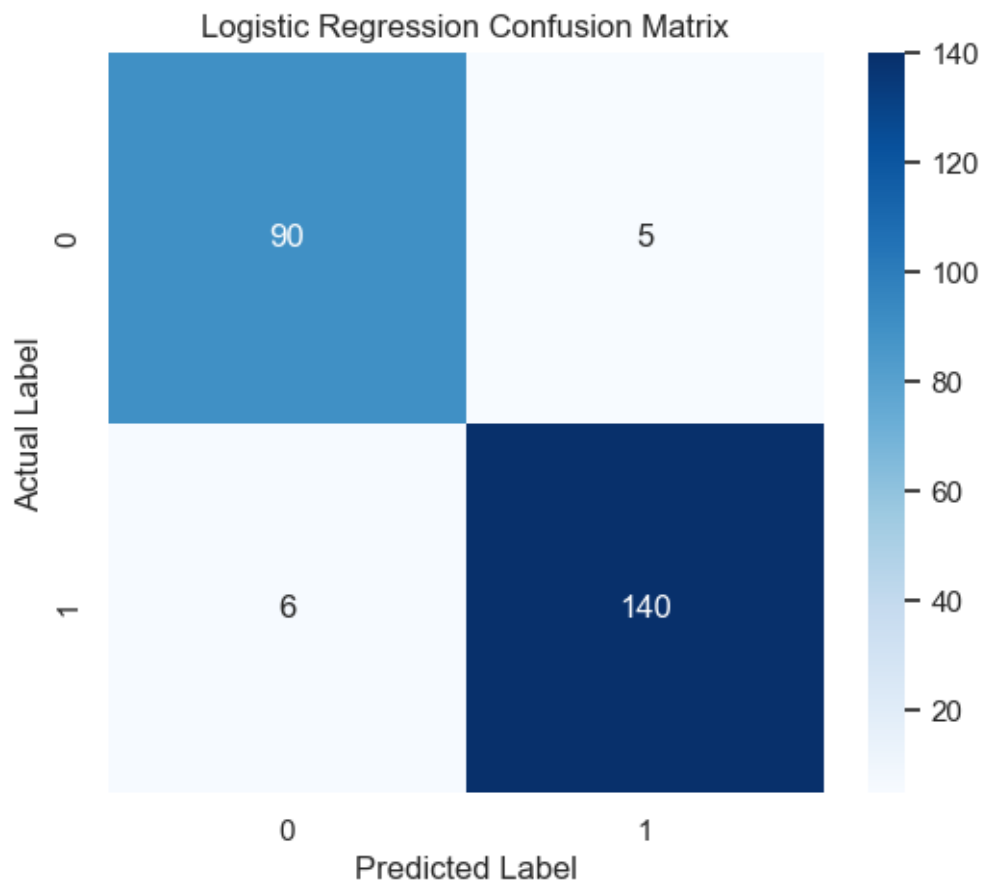
### Logistic Regression Results

The Logistic Regression model demonstrated strong classification performance on the maternal health dataset.

The evaluation metrics showed:

- high accuracy,
- balanced precision and recall,
- and strong ROC-AUC performance.

The model successfully identified both low-risk and high-risk maternal cases with minimal classification error.



Logistic Regression confusion matrix showing classification performance for maternal risk prediction.

## 5.2 K-Nearest Neighbors (KNN)

K-Nearest Neighbors (KNN) is a supervised machine learning algorithm based on similarity and distance measurements.

Unlike Logistic Regression, KNN does not build an explicit mathematical model during training. Instead, it classifies new observations based on the nearest neighboring data points.

In this study, KNN was applied to classify maternal patients into low-risk and high-risk categories using medical feature similarity.

### KNN Theory

KNN predicts the class of a sample by analyzing the classes of its nearest neighbors.

The distance between observations is commonly calculated using Euclidean Distance:

$$d(x, y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$$

where:

- $x_i$  and  $y_i$  represent feature values of two observations,
- and  $n$  represents the number of features.

The algorithm:

1. calculates distances between observations,
2. selects the nearest  $k$  neighbors,
3. and assigns the majority class.

### Why KNN Was Used

KNN was selected because:

- it is simple and intuitive,
- performs effectively on structured datasets,
- and captures local similarity patterns between maternal health records.

The algorithm is highly sensitive to feature scale; therefore, feature standardization using StandardScaler was essential before training.

KNN is particularly useful when:

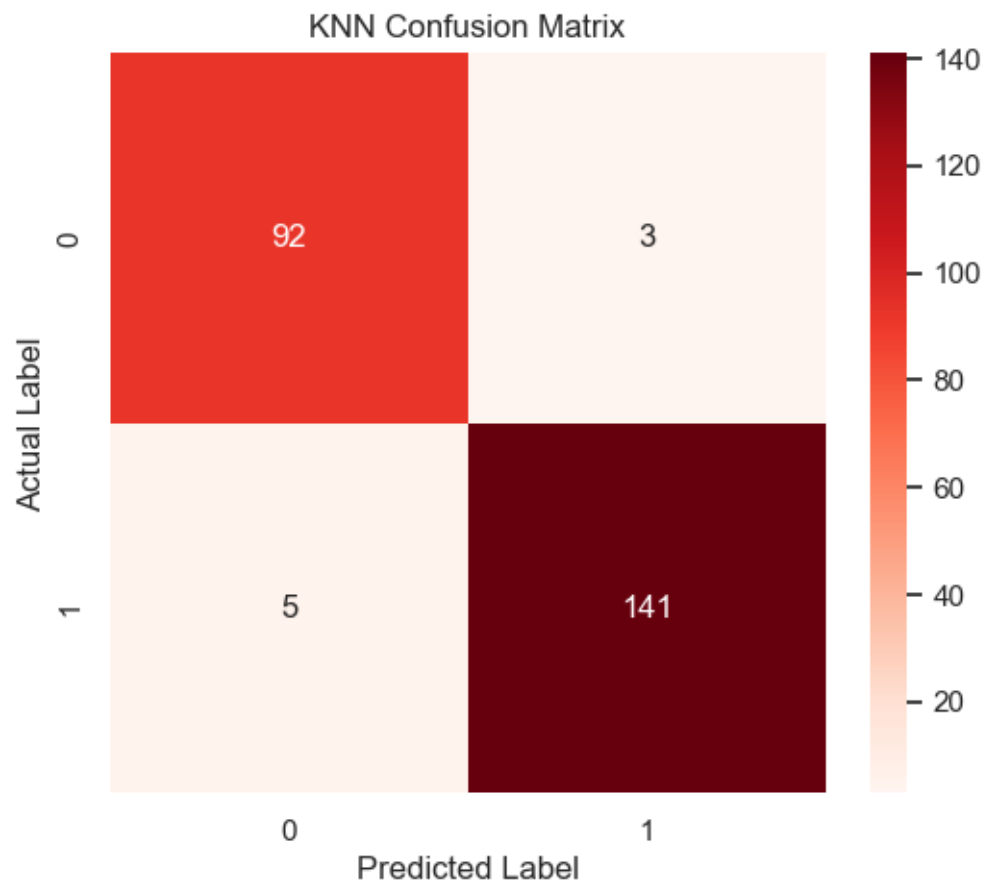
- similar medical profiles correspond to similar risk categories.

### KNN Results

The KNN model achieved strong prediction performance with:

- high classification accuracy,
- excellent ROC-AUC score,
- and balanced precision-recall values.

The confusion matrix demonstrated that KNN successfully classified most maternal health observations with only a small number of misclassifications.



KNN confusion matrix demonstrating maternal risk classification performance.

### 5.3 Support Vector Machine (SVM)

Support Vector Machine (SVM) is a powerful classification algorithm designed to identify the optimal decision boundary between classes.

SVM attempts to maximize the margin separating different categories while minimizing classification errors.

In this project, SVM was used to separate:

- low-risk maternal cases,
- and high-risk maternal cases.

#### SVM Theory

SVM constructs a hyperplane that maximizes the separation margin between classes.

The decision boundary can be expressed as:

$$w^T x + b = 0$$

where:

- $w$  represents feature weights,
- $x$  represents feature vectors,
- and  $b$  represents bias.

The optimization objective is to maximize the margin:

$$\text{Margin} = \frac{2}{\|w\|}$$

A larger margin generally improves model generalization.

### Why SVM Was Used

SVM was selected because:

- it performs strongly on high-dimensional datasets,
- handles binary classification effectively,
- and provides strong generalization capability.

SVM is particularly effective when:

- the boundary between classes is complex,
- and accurate separation is required.

Feature scaling was especially important for SVM because the algorithm relies heavily on geometric distances.

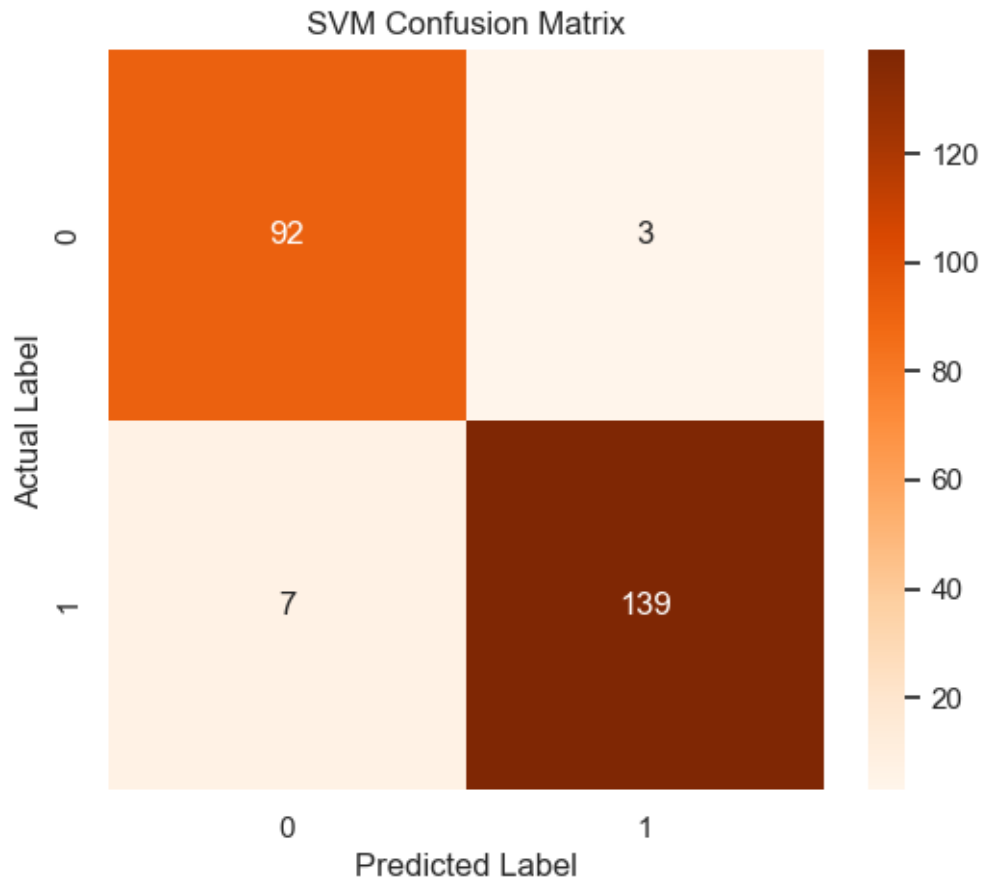
### SVM Results

The SVM model achieved:

- high accuracy,
- strong ROC-AUC performance,
- and stable cross-validation results.

The model demonstrated effective separation between low-risk and high-risk maternal patients with relatively few classification errors.





SVM confusion matrix illustrating classification capability on maternal health data.

#### 5.4 Decision Tree

Decision Tree is a rule-based machine learning algorithm that classifies observations through hierarchical decision structures.

The model recursively splits the dataset into smaller subsets using feature-based conditions.

In this study, Decision Tree was applied to classify maternal risk levels based on medical indicators.

##### Decision Tree Theory

Decision Trees select splitting rules that maximize information gain and reduce impurity.

One commonly used impurity metric is Gini Impurity:

$$Gini = 1 - \sum_{i=1}^c p_i^2$$

where:

- $p_i$  represents the probability of class  $i$ ,
- and  $c$  is the number of classes.

The algorithm recursively partitions the data until:

- leaf nodes become sufficiently pure,
- or stopping criteria are reached.

### Why Decision Tree Was Used

Decision Trees were selected because:

- they are highly interpretable,
- capable of capturing nonlinear relationships,
- and require minimal data assumptions.

In medical datasets, tree-based models are useful because:

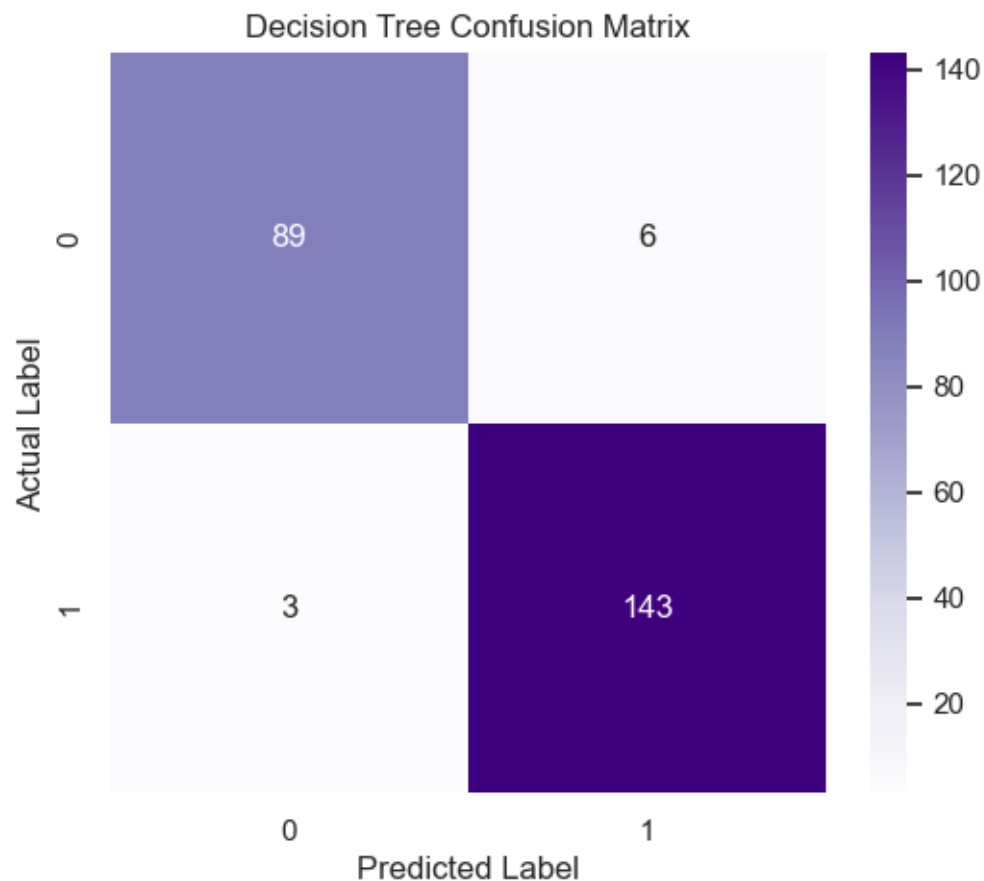
- decision paths can be interpreted clinically,
- and important features can be identified visually.

### Decision Tree Results

The Decision Tree model demonstrated strong classification performance with:

- high accuracy,
- high recall values,
- and effective risk separation.

The model successfully identified high-risk maternal cases while maintaining strong overall prediction consistency.



Decision Tree confusion matrix for maternal risk classification.

## 5.5 Random Forest

Random Forest is an ensemble learning algorithm that combines multiple decision trees to improve prediction accuracy and reduce overfitting.

Instead of relying on a single tree, Random Forest aggregates predictions from many independent trees.

In this project, Random Forest achieved the strongest overall classification performance.

### Random Forest Theory

Random Forest creates multiple decision trees using:

- bootstrap sampling,
- and random feature selection.

Final predictions are determined using majority voting:

$$\hat{y} = \text{mode}(T_1(x), T_2(x), \dots, T_n(x))$$

where:

- $T_i(x)$  represents predictions from individual trees.

This ensemble approach reduces:

- variance,
- overfitting,
- and instability.

### Why Random Forest Was Used

Random Forest was selected because:

- it performs exceptionally well on structured tabular datasets,
- handles nonlinear relationships effectively,
- and improves generalization performance.

The algorithm also provides:

- feature importance analysis,
- strong robustness,
- and stable classification performance.

These characteristics make Random Forest highly suitable for medical risk prediction tasks.

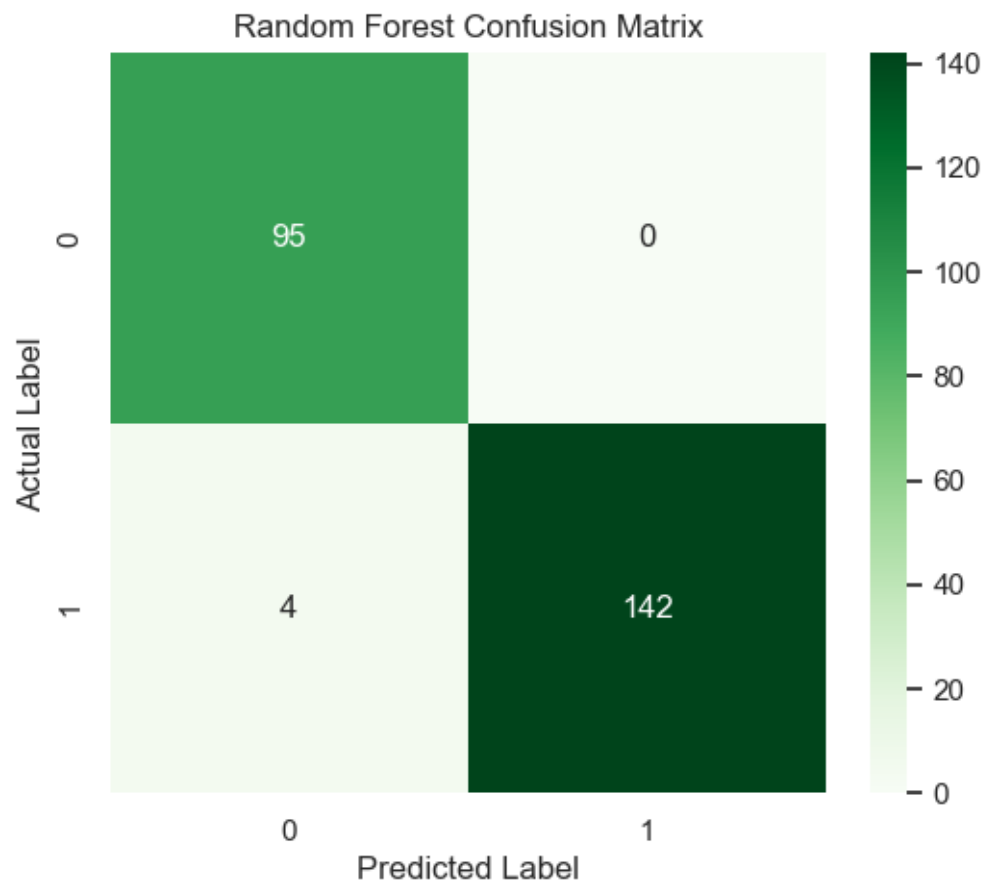
### Random Forest Results

Random Forest achieved the best overall performance among all evaluated models.

The model demonstrated:

- the highest accuracy,
- strongest ROC-AUC score,
- and strongest cross-validation consistency.

The confusion matrix showed very few misclassifications, indicating highly reliable maternal risk prediction capability.



Random Forest confusion matrix showing the highest-performing classification results.

## 6. Model Evaluation Metrics

Evaluating machine learning models is a critical stage in classification studies because a single metric alone cannot fully represent model performance.

In medical prediction systems, evaluation becomes even more important because:

- incorrect classifications may lead to severe clinical consequences,
- and different error types affect healthcare decisions differently.

For this reason, multiple evaluation metrics were used in this project to comprehensively analyze model performance.

The following metrics were applied:

- Accuracy
- Confusion Matrix
- Precision
- Recall
- F1-Score
- ROC-AUC

- Cross Validation

These metrics collectively provide:

- classification quality,
- prediction reliability,
- generalization capability,
- and robustness analysis.

## 6.1 Accuracy

Accuracy measures the proportion of correctly classified observations among all predictions.

It is one of the most commonly used evaluation metrics in classification tasks.

The mathematical definition of accuracy is:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

where:

- $TP$  = True Positives
- $TN$  = True Negatives
- $FP$  = False Positives
- $FN$  = False Negatives

Accuracy indicates the overall prediction success of the model.

For example:

- an accuracy of 0.98 means the model correctly classified 98% of all maternal health observations.

However, accuracy alone may become misleading in imbalanced datasets because:

- a model can achieve high accuracy while still failing minority classes.

Therefore, additional metrics were also analyzed.

## 6.2 Confusion Matrix

The confusion matrix provides a detailed breakdown of classification results.

Unlike accuracy, it reveals:

- which classes were predicted correctly,
- and which classes were misclassified.

A binary classification confusion matrix is represented as:

$$\begin{bmatrix} TN & FP \\ FN & TP \end{bmatrix}$$

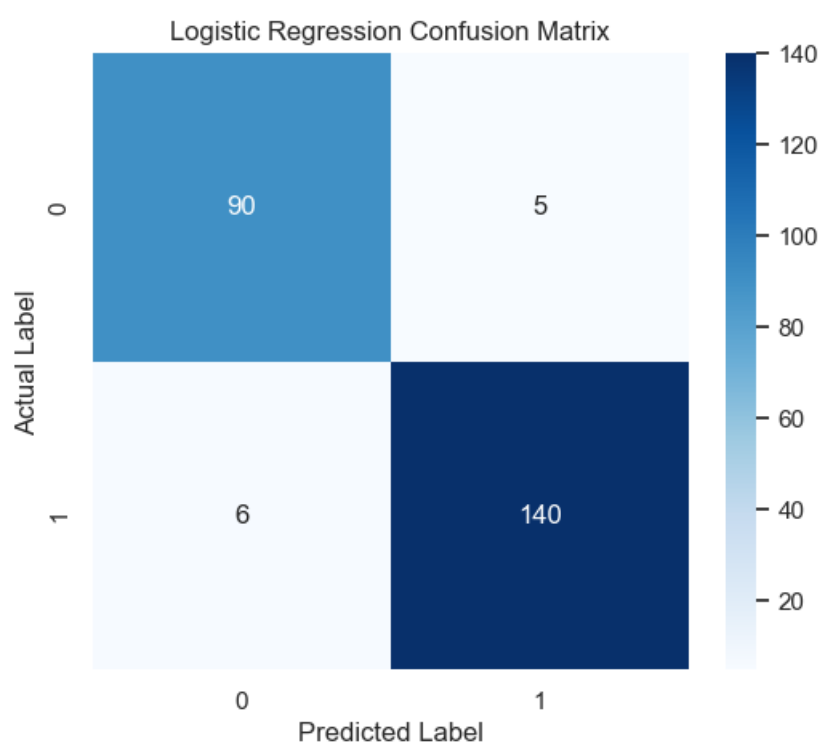
The matrix components represent:

| <b>Component</b>           | <b>Meaning</b>                      |
|----------------------------|-------------------------------------|
| <i>True Positive (TP)</i>  | High-risk cases correctly predicted |
| <i>True Negative (TN)</i>  | Low-risk cases correctly predicted  |
| <i>False Positive (FP)</i> | Low-risk predicted as high-risk     |
| <i>False Negative (FN)</i> | High-risk predicted as low-risk     |

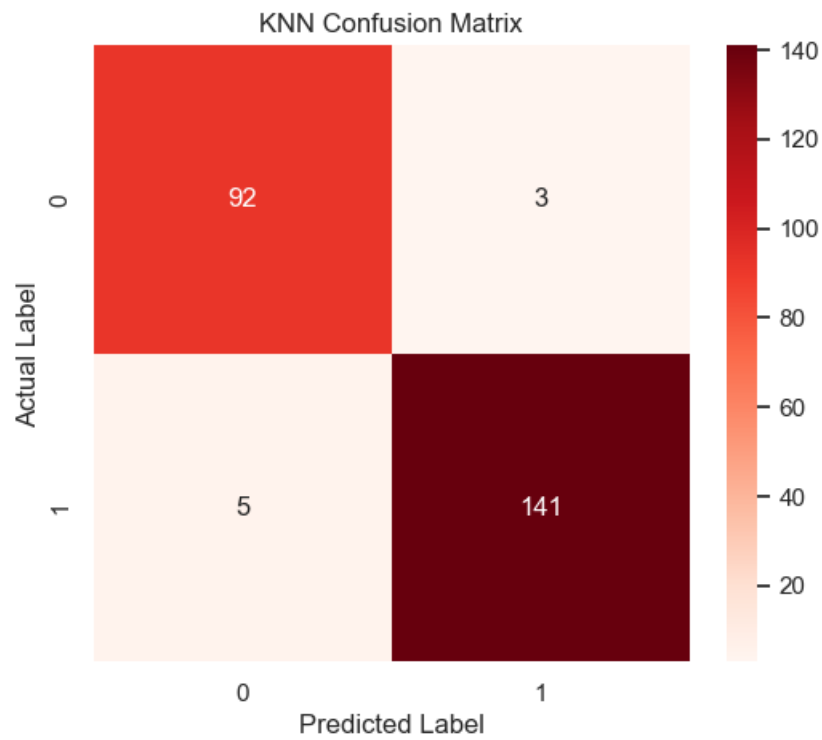
In medical prediction systems:

- False Negatives are particularly dangerous because high-risk patients may remain undetected.

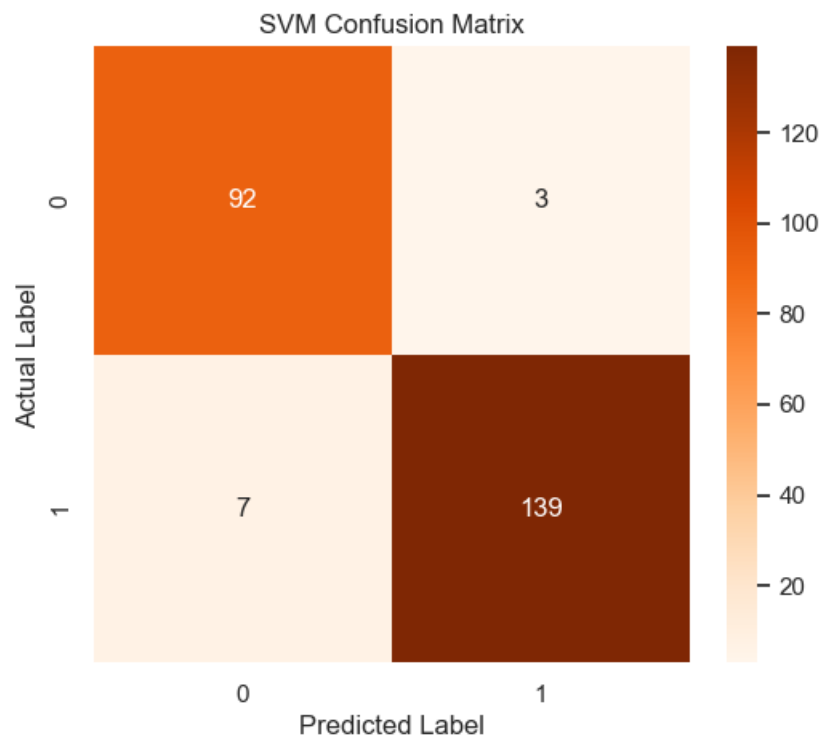
Confusion matrices were generated for all machine learning models in this study.



Logistic Regression confusion matrix.

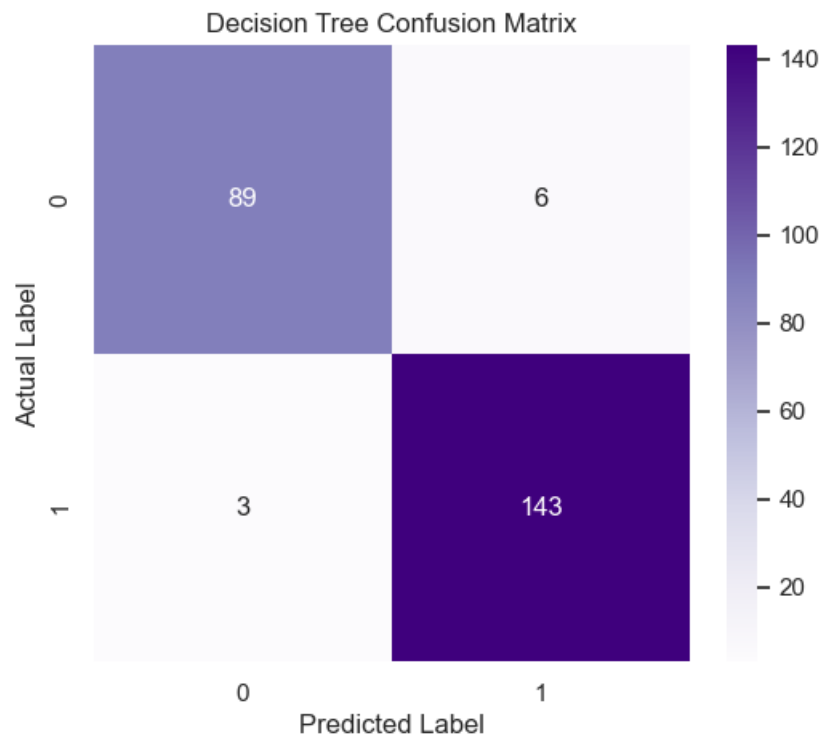


KNN confusion matrix.

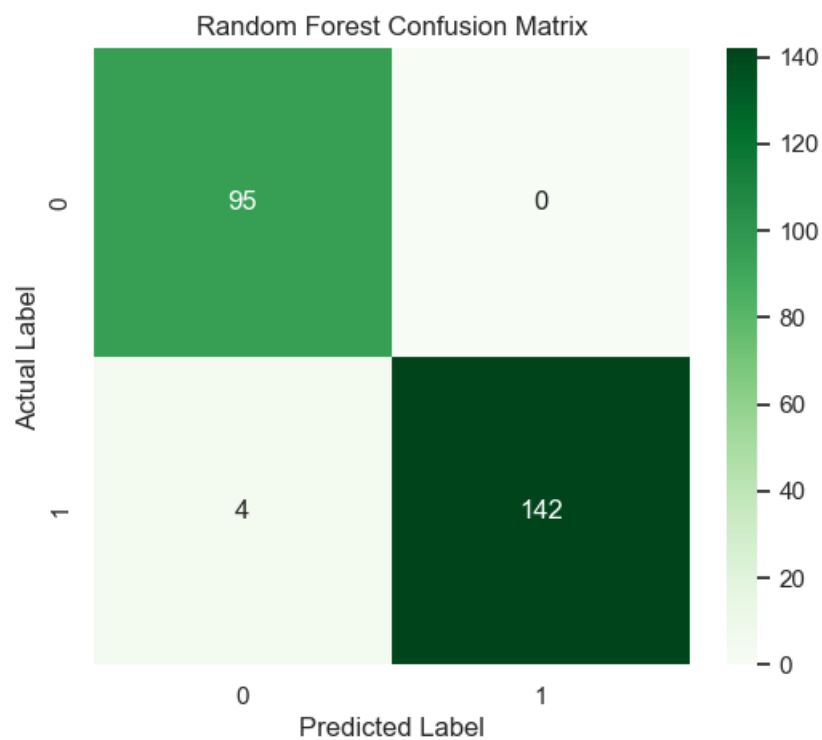


SVM confusion matrix.





Decision Tree confusion matrix.



Random Forest confusion matrix.

### 6.3 Precision

Precision measures how many positively predicted observations were actually positive.

The mathematical definition is:

$$Precision = \frac{TP}{TP + FP}$$

A high precision score indicates:

- fewer false positive predictions,
- and more reliable positive classifications.

In maternal health prediction:

- high precision means patients classified as high-risk are likely to truly belong to the high-risk category.

## 6.4 Recall

Recall measures how many actual positive observations were successfully identified.

The formula is:

$$Recall = \frac{TP}{TP + FN}$$

Recall is critically important in healthcare applications because:

- missing high-risk patients may lead to severe medical consequences.

A high recall value indicates:

- the model successfully detects most high-risk maternal cases.

## 6.5 F1-Score

F1-score combines precision and recall into a single metric.

It is especially useful when:

- class distributions are imbalanced,
- or both false positives and false negatives are important.

The F1-score formula is:

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

The metric represents the harmonic mean of:

- precision,
- and recall.

A high F1-score indicates:

- balanced classification capability,
- and stable predictive performance.

## 6.6 ROC-AUC

ROC-AUC evaluates the model's ability to distinguish between classes.

ROC stands for:

- Receiver Operating Characteristic.

AUC stands for:

- Area Under the Curve.

The ROC curve plots:

- True Positive Rate (TPR)  
against
- False Positive Rate (FPR).

The curve is represented using:

$$TPR = \frac{TP}{TP + FN}$$

and

$$FPR = \frac{FP}{FP + TN}$$

The AUC value ranges between:

- 0 and 1.

Interpretation:

| <b><i>AUC Value</i></b> | <b><i>Interpretation</i></b> |
|-------------------------|------------------------------|
| <i>0.50</i>             | Random classification        |
| <i>0.70 – 0.80</i>      | Acceptable                   |
| <i>0.80 – 0.90</i>      | Excellent                    |
| <i>&gt; 0.90</i>        | Outstanding                  |

In this project, all machine learning models achieved ROC-AUC scores above 0.95, indicating outstanding classification capability.

## 6.7 Cross Validation

Cross Validation evaluates how well a machine learning model generalizes to unseen data.

Instead of relying on a single train-test split, the dataset is repeatedly partitioned into multiple folds.

In this project:

- 5-Fold Cross Validation was applied.

The process can be summarized as:

1. The dataset is divided into 5 subsets.
2. Four subsets are used for training.
3. One subset is used for validation.
4. The process repeats until every subset has been used for testing.

The final score is computed as:

$$CV_{mean} = \frac{1}{k} \sum_{i=1}^k Score_i$$

where:

- $k$  is the number of folds.

Cross Validation provides:

- more reliable performance estimation,
- reduced overfitting risk,
- and stronger statistical confidence.

Models with:

- high cross-validation mean,
- and low variation across folds

are considered more stable and generalizable.

## Evaluation Summary

The evaluation metrics collectively demonstrated that all machine learning models achieved strong maternal risk classification performance.

The applied metrics provided:

- prediction quality analysis,
- class discrimination capability,
- robustness evaluation,
- and generalization assessment.

Among all evaluated algorithms:

- Random Forest achieved the strongest overall performance,
- followed closely by Decision Tree, SVM, and KNN.

The evaluation results confirmed that machine learning algorithms can effectively classify maternal health risk using medical indicators and clinical measurements.

## 7. Experimental Results

This section presents the experimental performance results of all machine learning models applied to the maternal health risk prediction dataset.

Each model was evaluated using:

- Accuracy,
- ROC-AUC,
- Confusion Matrix,
- Precision,
- Recall,
- and F1-score metrics.

The objective of this analysis was to:

- compare model effectiveness,
- identify the strongest classifier,
- and evaluate the reliability of maternal risk prediction.

The evaluated models include:

1. Logistic Regression
2. K-Nearest Neighbors (KNN)
3. Support Vector Machine (SVM)
4. Decision Tree
5. Random Forest

The experimental findings demonstrate that all models achieved high classification performance, while ensemble-based approaches provided the strongest overall results.

7.1 Logistic Regression Results

The Logistic Regression model demonstrated strong binary classification capability on the maternal health dataset.

The model achieved:

- high overall accuracy,
- balanced precision and recall,
- and strong ROC-AUC performance.

Logistic Regression Performance Metrics

| Metric    | Value   |
|-----------|---------|
| Accuracy  | 95.44 % |
| ROC-AUC   | 0.9531  |
| Precision | 0.97    |
| Recall    | 0.96    |
| F1-Score  | 0.96    |

---

Confusion Matrix Analysis

The confusion matrix revealed:

- 90 correctly classified low-risk cases,
- 140 correctly classified high-risk cases,
- and only 11 total misclassifications.

This indicates that Logistic Regression successfully learned the underlying relationship between maternal health indicators and risk categories.

Logistic Regression Observation

The Logistic Regression model achieved reliable and stable classification performance despite being one of the simplest algorithms used in this study.

The model demonstrated:

- strong probability-based classification capability,
- balanced prediction behavior,
- and strong generalization performance.

Its relatively low computational complexity also makes it suitable for real-time medical screening systems.

7.2 K-Nearest Neighbors (KNN) Results

The KNN model demonstrated excellent classification capability by utilizing similarity-based learning between maternal health observations.

The algorithm successfully captured local feature patterns within the dataset.

KNN Performance Metrics

| <i><b>Metric</b></i> | <i><b>Value</b></i> |
|----------------------|---------------------|
| <i>Accuracy</i>      | 96.68 %             |
| <i>ROC-AUC</i>       | 0.9671              |
| <i>Precision</i>     | 0.98                |
| <i>Recall</i>        | 0.97                |
| <i>F1-Score</i>      | 0.97                |

Confusion Matrix Analysis

The KNN confusion matrix showed:

- 92 correctly classified low-risk cases,
- 141 correctly classified high-risk cases,
- and only 8 misclassifications.

The model demonstrated particularly strong classification performance after feature scaling.

KNN Observation

KNN achieved highly competitive classification results with:

- excellent sensitivity,
- strong ROC-AUC performance,
- and balanced prediction capability.

The model effectively utilized neighborhood similarity among maternal health profiles and demonstrated strong learning capability after standardization.

7.3 Support Vector Machine (SVM) Results

The Support Vector Machine model produced highly accurate classification results by maximizing the decision boundary margin between maternal risk categories.

The model demonstrated strong separation capability between:

- low-risk,
- and high-risk maternal cases.

SVM Performance Metrics

| <i><b>Metric</b></i> | <i><b>Value</b></i> |
|----------------------|---------------------|
| <i>Accuracy</i>      | 95.85 %             |
| <i>ROC-AUC</i>       | 0.9602              |
| <i>Precision</i>     | 0.98                |
| <i>Recall</i>        | 0.95                |
| <i>F1-Score</i>      | 0.97                |

Confusion Matrix Analysis

The confusion matrix showed:

- 92 correctly classified low-risk cases,
- 139 correctly classified high-risk cases,
- and only 10 total classification errors.

The SVM model successfully minimized overlap between risk categories.

SVM Observation

SVM demonstrated:

- strong margin-based learning capability,
- high robustness,
- and excellent generalization performance.

The model effectively separated maternal health risk groups while maintaining low misclassification rates.

Its high ROC-AUC value further confirmed strong class discrimination capability.

7.4 Decision Tree Results

The Decision Tree model achieved strong classification performance through hierarchical decision rules based on maternal health indicators.

The algorithm successfully captured nonlinear relationships between clinical variables and maternal risk levels.

Decision Tree Performance Metrics

| <i><b>Metric</b></i> | <i><b>Value</b></i> |
|----------------------|---------------------|
| <i>Accuracy</i>      | 96.27 %             |
| <i>ROC-AUC</i>       | 0.9581              |
| <i>Precision</i>     | 0.96                |
| <i>Recall</i>        | 0.98                |
| <i>F1-Score</i>      | 0.97                |

---



Confusion Matrix Analysis

The Decision Tree confusion matrix demonstrated:

- 89 correctly classified low-risk cases,
- 143 correctly classified high-risk cases,
- and only 9 classification errors.

The model achieved particularly strong recall performance for high-risk maternal patients.

Decision Tree Observation

The Decision Tree model demonstrated:

- strong interpretability,
- high recall performance,
- and successful nonlinear classification capability.

The model effectively identified high-risk maternal cases while maintaining overall prediction consistency.

Its transparent decision structure also improves explainability in medical applications.

7.5 Random Forest Results

The Random Forest model achieved the strongest overall performance among all evaluated machine learning algorithms.

The ensemble learning structure significantly improved:

- prediction stability,
- classification accuracy,
- and generalization performance.

Random Forest Performance Metrics

| <i><b>Metric</b></i> | <i><b>Value</b></i> |
|----------------------|---------------------|
| <i>Accuracy</i>      | 98.34 %             |
| <i>ROC-AUC</i>       | 0.9863              |
| <i>Precision</i>     | 1.00                |
| <i>Recall</i>        | 0.97                |
| <i>F1-Score</i>      | 0.99                |

Confusion Matrix Analysis

The Random Forest confusion matrix showed:

- 95 correctly classified low-risk cases,

- 142 correctly classified high-risk cases,
- and only 4 total misclassifications.

The model achieved:

- the lowest error rate,
- the highest accuracy,
- and the strongest overall classification consistency.

### Random Forest Observation

Random Forest outperformed all other machine learning models in this study.

The model demonstrated:

- exceptional classification capability,
- strong robustness,
- high ROC-AUC performance,
- and minimal prediction error.

The ensemble-based architecture improved generalization and reduced overfitting risk, making Random Forest the most reliable model for maternal risk prediction in this project.

### Experimental Results Summary

The experimental findings demonstrated that all machine learning algorithms achieved strong classification performance on the maternal health dataset.

However:

- Random Forest achieved the highest overall performance,
- followed by KNN, Decision Tree, SVM, and Logistic Regression.

The results confirmed that machine learning methods can effectively identify maternal health risks using clinical and medical indicators with high predictive accuracy.

## 8. Cross Validation Analysis

Cross Validation is an essential evaluation technique used to measure how well a machine learning model generalizes to unseen data.

Unlike a single train-test split, Cross Validation repeatedly evaluates the model on different subsets of the dataset, providing:

- more reliable performance estimation,
- reduced evaluation bias,
- and stronger statistical confidence.

In this project, 5-Fold Cross Validation was applied to all machine learning algorithms.

The dataset was divided into five equal subsets:

- four subsets were used for training,
- while the remaining subset was used for validation.

This process was repeated five times until every subset had been used as a validation set once.

The final Cross Validation score was calculated using the mean performance across all folds.

The mathematical formulation is:

$$CV_{mean} = \frac{1}{k} \sum_{i=1}^k Score_i$$

where:

- $k$  represents the number of folds,
- and  $Score_i$  represents the validation score for each fold.

### 8.1 Logistic Regression Cross Validation

The Logistic Regression model produced highly stable validation results across all folds.

#### Logistic Regression CV Scores

| <b><i>Fold</i></b> | <b><i>Score</i></b> |
|--------------------|---------------------|
| <i>Fold 1</i>      | 0.9689              |
| <i>Fold 2</i>      | 0.9793              |
| <i>Fold 3</i>      | 0.9689              |
| <i>Fold 4</i>      | 0.9896              |
| <i>Fold 5</i>      | 0.9583              |

Mean CV Score

$$CV_{mean} = 0.9730$$

#### Logistic Regression Interpretation

The relatively small variation between folds indicates:

- stable learning behavior,
- strong generalization capability,
- and low overfitting risk.

The model maintained consistently high performance across multiple validation subsets.

## 8.2 KNN Cross Validation

The KNN model also demonstrated strong validation performance.

### KNN CV Scores

| <b><i>Fold</i></b> | <b><i>Score</i></b> |
|--------------------|---------------------|
| <i>Fold 1</i>      | 0.9585              |
| <i>Fold 2</i>      | 0.9845              |
| <i>Fold 3</i>      | 0.9689              |
| <i>Fold 4</i>      | 0.9689              |
| <i>Fold 5</i>      | 0.9583              |

### Mean CV Score

$$CV_{mean} = 0.9678$$

### KNN Interpretation

The KNN model demonstrated:

- strong prediction consistency,
- stable neighborhood learning,
- and effective generalization after feature scaling.

The fold results remained relatively balanced, indicating reliable model stability.

## 8.3 SVM Cross Validation

The SVM model achieved highly competitive validation performance.

### SVM CV Scores

| <b><i>Fold</i></b> | <b><i>Score</i></b> |
|--------------------|---------------------|
| <i>Fold 1</i>      | 0.9741              |
| <i>Fold 2</i>      | 0.9896              |
| <i>Fold 3</i>      | 0.9637              |
| <i>Fold 4</i>      | 0.9845              |
| <i>Fold 5</i>      | 0.9531              |

### Mean CV Score

$$CV_{mean} = 0.9730$$

### SVM Interpretation

SVM demonstrated:

- excellent margin-based learning stability,
- strong robustness,
- and highly consistent validation performance.

The model generalized effectively across multiple validation folds with minimal performance fluctuation.

## 8.4 Decision Tree Cross Validation

The Decision Tree model achieved strong validation performance but showed slightly higher fold variability compared to ensemble-based methods.

### Decision Tree CV Scores

| <b><i>Fold</i></b> | <b><i>Score</i></b> |
|--------------------|---------------------|
| <i>Fold 1</i>      | 0.9482              |
| <i>Fold 2</i>      | 0.9741              |
| <i>Fold 3</i>      | 0.9689              |
| <i>Fold 4</i>      | 0.9482              |
| <i>Fold 5</i>      | 0.9583              |

Mean CV Score

$$CV_{mean} = 0.9595$$

### Decision Tree Interpretation

The Decision Tree model demonstrated:

- strong classification capability,
- but slightly lower stability compared to Random Forest and SVM.

This behavior is expected because single decision trees are more sensitive to training data variations and may overfit specific subsets.

## 8.5 Random Forest Cross Validation

Random Forest achieved the strongest Cross Validation performance among all evaluated models.

### Random Forest CV Scores

| <b><i>Fold</i></b> | <b><i>Score</i></b> |
|--------------------|---------------------|
| <i>Fold 1</i>      | 0.9793              |
| <i>Fold 2</i>      | 0.9948              |
| <i>Fold 3</i>      | 0.9845              |
| <i>Fold 4</i>      | 0.9896              |
| <i>Fold 5</i>      | 0.9635              |

Mean CV Score

$$CV_{mean} = 0.9823$$

### Random Forest Interpretation

Random Forest demonstrated:

- outstanding generalization capability,
- highly stable validation performance,
- and minimal fold variation.

The ensemble learning structure significantly improved robustness by combining predictions from multiple decision trees.

The consistently high validation scores indicate:

- low overfitting risk,
- strong predictive reliability,
- and excellent adaptability to unseen maternal health data.

### Cross Validation Summary

Cross Validation analysis confirmed that all machine learning models generalized effectively to unseen data.

Among all algorithms:

- Random Forest achieved the highest validation performance,
- followed by Logistic Regression and SVM.

The results demonstrated that:

- ensemble learning methods provide superior stability,
- while simpler models also maintain strong predictive capability.

Overall, the Cross Validation findings confirmed that the developed machine learning pipeline is:

- statistically reliable,
- stable,
- and suitable for maternal health risk prediction applications.

## 9. Feature Importance Analysis

Feature importance analysis is a critical component of machine learning interpretability because it helps identify which variables contribute most to prediction performance.

In healthcare-related machine learning systems, interpretability is especially important because:

- clinicians need to understand why predictions are generated,
- and medically influential variables must be identifiable.

In this project, feature importance analysis was performed using the Random Forest model because:

- Random Forest naturally provides feature importance scores,
- and it achieved the best overall classification performance.

The importance values indicate how strongly each medical feature contributes to maternal risk prediction.

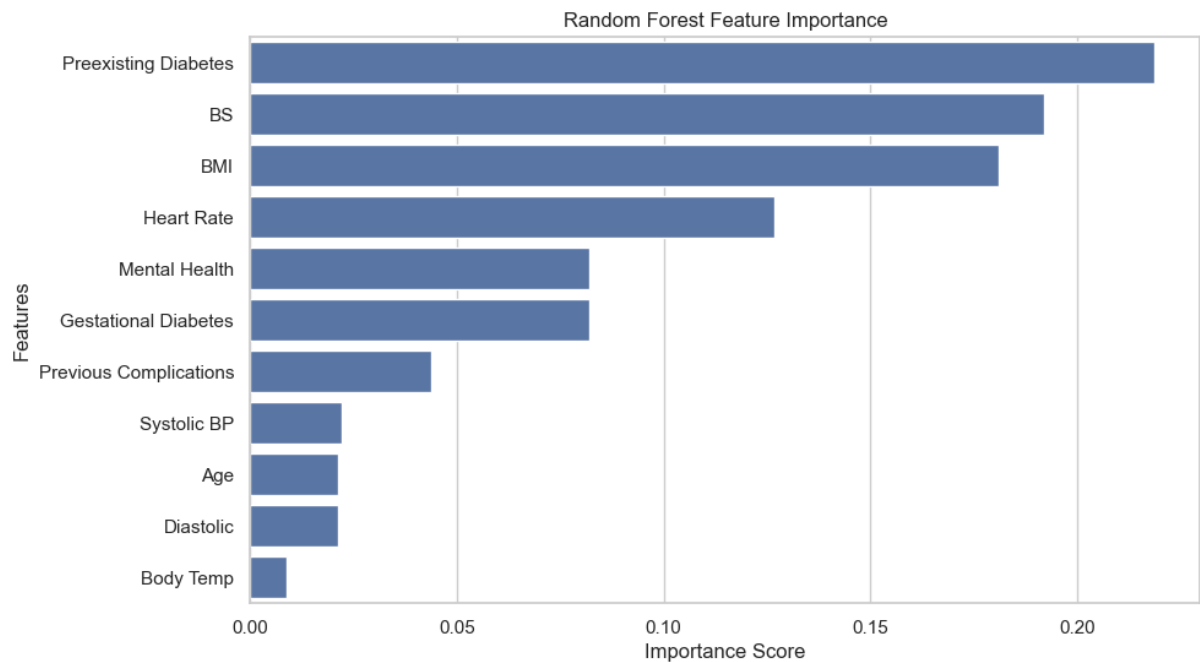
Higher importance scores correspond to:

- stronger predictive influence,
- greater contribution to decision-making,
- and higher impact on classification performance.

9.1 Random Forest Feature Importance

The Random Forest model calculated feature importance values using decision-tree-based split contributions.

The importance ranking demonstrated that some medical indicators had significantly stronger influence on maternal risk prediction.



Random Forest feature importance analysis showing the contribution strength of maternal health indicators.

9.2 Top Important Features

The top-ranked features identified by the Random Forest model are presented below.

| Feature              | Importance Score |
|----------------------|------------------|
| Preexisting Diabetes | 0.2189           |
| BS (Blood Sugar)     | 0.1920           |
| BMI                  | 0.1813           |
| Heart Rate           | 0.1268           |

The results indicate that:

- diabetes-related variables,
- metabolic indicators,
- and cardiovascular measurements

play the most significant role in maternal risk prediction.

### 9.3 Medical Interpretation of Important Features

The feature importance findings align strongly with real-world medical knowledge regarding maternal health complications.

The analysis demonstrated that maternal risk prediction is heavily influenced by:

- diabetic conditions,
- blood sugar abnormalities,
- and obesity-related factors.

#### Preexisting Diabetes

Preexisting Diabetes achieved the highest importance score:

$$Importance = 0.2189$$

This result is medically meaningful because diabetes is one of the strongest risk factors during pregnancy.

Preexisting diabetes may lead to:

- hypertension,
- fetal complications,
- cardiovascular stress,
- and pregnancy-related metabolic disorders.

The high importance value indicates that:

- maternal diabetes history strongly influences machine learning predictions,
- and plays a major role in identifying high-risk pregnancies.

#### Blood Sugar (BS)

Blood Sugar (BS) was identified as the second most important feature:

$$Importance = 0.1920$$



Blood Sugar is a critical clinical indicator because abnormal glucose levels are strongly associated with:

- gestational diabetes,
- insulin resistance,
- and maternal complications.

Elevated blood sugar values may increase:

- maternal health risks,
- pregnancy complications,
- and emergency intervention probability.

The model strongly relied on blood sugar levels when distinguishing between:

- low-risk,
- and high-risk maternal cases.

### Body Mass Index (BMI)

BMI achieved the third highest importance score:

$$Importance = 0.1813$$

BMI is an important maternal health indicator because obesity is strongly associated with:

- gestational hypertension,
- diabetes,
- cardiovascular stress,
- and pregnancy complications.

Higher BMI values may increase the likelihood of:

- surgical delivery,
- fetal complications,
- and maternal metabolic disorders.

The model identified BMI as one of the strongest predictive indicators for maternal risk classification.

### Heart Rate

Heart Rate also demonstrated relatively high importance:

$$Importance = 0.1268$$

Abnormal heart rate measurements may indicate:

- cardiovascular stress,
- hypertension,
- or systemic physiological complications during pregnancy.

The importance score suggests that cardiovascular measurements contribute significantly to maternal risk prediction.

## Mental Health

Mental Health achieved a moderate importance score:

$$\text{Importance} = 0.0819$$

Although lower than diabetes-related variables, mental health conditions may still influence:

- stress response,
- hormonal balance,
- and pregnancy outcomes.

The findings suggest that psychological health contributes to maternal risk assessment alongside physical health indicators.

## 9.4 Low-Importance Features

Some variables demonstrated relatively low importance scores, including:

- Body Temperature,
- Systolic Blood Pressure,
- Diastolic Blood Pressure,
- and Age.

This does not necessarily mean these variables are medically unimportant.

Instead, it suggests that:

- within this specific dataset,
- these variables contributed less to model decision-making compared to stronger predictors such as diabetes and blood sugar.

## 9.5 Interpretation Summary

Feature importance analysis demonstrated that:

- diabetes-related indicators,
- metabolic measurements,
- and obesity-related variables

are the strongest predictors of maternal health risk.

The findings align closely with established medical knowledge and confirm that the machine learning models successfully captured clinically meaningful relationships within the dataset.

The Random Forest model not only achieved the highest classification performance but also provided strong interpretability by identifying medically relevant predictive factors.

This interpretability significantly improves:

- model transparency,
- clinical trustworthiness,
- and real-world applicability in maternal healthcare prediction systems.

## 10. Final Model Comparison

This section presents the final comparative analysis of all machine learning models implemented in this study.

The objective of this comparison is to:

- evaluate overall model performance,
- identify the strongest classification algorithm,
- and determine the most reliable model for maternal health risk prediction.

The comparison was performed using:

- Accuracy,
- ROC-AUC,
- and Cross Validation Mean scores.

These metrics collectively provide information regarding:

- prediction capability,
- classification reliability,
- model robustness,
- and generalization performance.

The evaluated machine learning algorithms include:

1. Logistic Regression
2. K-Nearest Neighbors (KNN)
3. Support Vector Machine (SVM)
4. Decision Tree
5. Random Forest

### 10.1 Final Performance Comparison Table

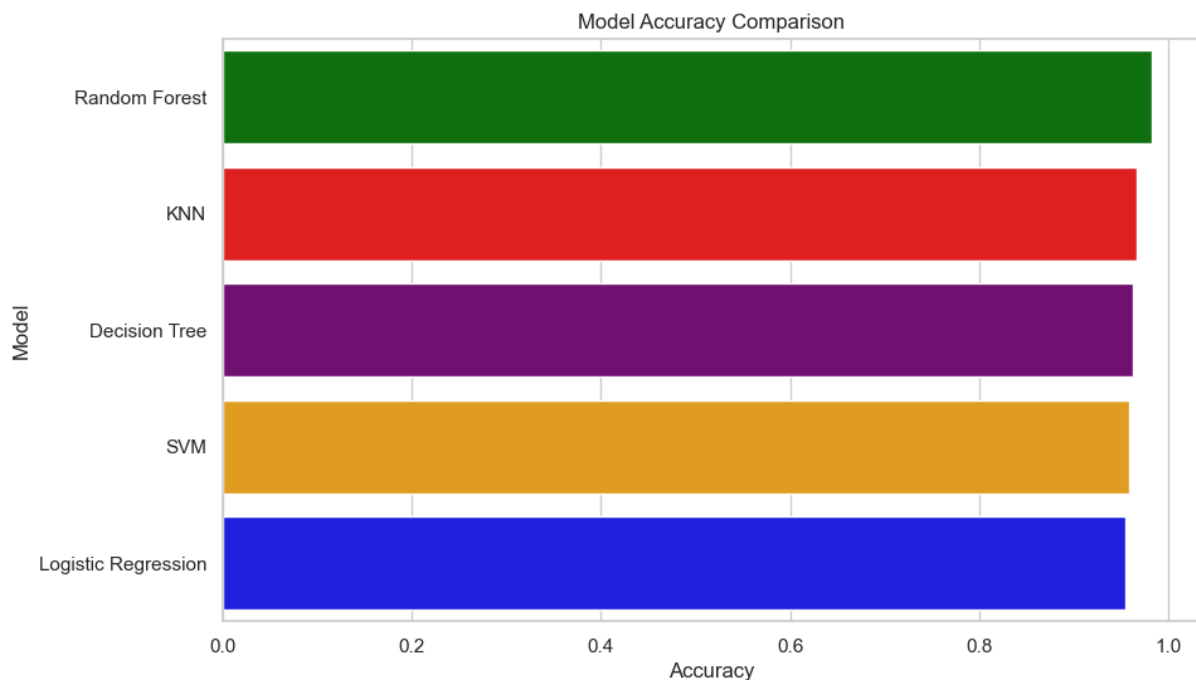
The final performance results of all machine learning models are summarized below.

| <i>Model</i>               | <i>Accuracy</i> | <i>ROC-AUC</i> | <i>CV Mean</i> |
|----------------------------|-----------------|----------------|----------------|
| <i>Random Forest</i>       | 98.34%          | 98.63%         | 98.23%         |
| <i>KNN</i>                 | 96.68%          | 96.71%         | 96.78%         |
| <i>Decision Tree</i>       | 96.27%          | 95.81%         | 95.95%         |
| <i>SVM</i>                 | 95.85%          | 96.02%         | 97.30%         |
| <i>Logistic Regression</i> | 95.44%          | 95.31%         | 97.30%         |

The results indicate that all models achieved strong classification performance.

However:

- ensemble and nonlinear methods generally outperformed simpler linear approaches.



Accuracy comparison graph of all machine learning models used for maternal risk prediction.

## 10.2 Accuracy Comparison Analysis

The accuracy comparison graph demonstrates that:

- Random Forest achieved the highest overall accuracy,
- while KNN and Decision Tree also produced highly competitive results.

The classification performances can be ranked as:

*Random Forest > KNN > Decision Tree > SVM > Logistic Regression*

Although Logistic Regression achieved slightly lower accuracy compared to other models, its performance remained highly competitive considering its simplicity and interpretability.

The relatively close performance values between models suggest that:

- the dataset contains highly informative medical features,
- and maternal risk patterns are strongly learnable using machine learning approaches.

### 10.3 ROC-AUC Comparison

ROC-AUC analysis demonstrated that all models achieved outstanding discrimination capability.

The highest ROC-AUC score was obtained by Random Forest:

$$ROC - AUC = 0.9863$$

This indicates:

- excellent separation between low-risk and high-risk maternal patients,
- and highly reliable probability-based classification performance.

All evaluated models achieved ROC-AUC scores above 0.95, which is considered outstanding in classification analysis.

### 10.4 Cross Validation Comparison

Cross Validation results revealed the generalization capability of each model.

Random Forest achieved the highest Cross Validation Mean:

$$CV_{mean} = 0.9823$$

The high and stable validation performance indicates:

- strong robustness,
- low overfitting risk,
- and excellent predictive stability.

Decision Tree demonstrated slightly higher fold variability due to the nature of single-tree learning.

Meanwhile:

- Logistic Regression,
- SVM,
- and KNN

also maintained highly stable validation performance across folds.

## 10.5 Best Model Selection

Based on:

- Accuracy,
- ROC-AUC,
- Cross Validation,
- and overall classification consistency,

the Random Forest model was selected as the best-performing machine learning algorithm in this project.

The model achieved:

- the highest classification accuracy,
- strongest ROC-AUC score,
- and strongest generalization capability.

The ensemble learning structure significantly improved:

- robustness,
- stability,
- and resistance to overfitting.

Additionally, Random Forest provided:

- feature importance analysis,
- strong interpretability,
- and reliable classification behavior on unseen maternal health data.

The final selected model can therefore be considered highly suitable for:

- maternal health risk prediction,
- clinical decision-support systems,
- and healthcare-oriented machine learning applications.

## 10.6 Final Comparison Summary

The final comparative analysis demonstrated that:

- all machine learning algorithms achieved strong classification capability,
- while ensemble learning approaches produced superior performance.

Among all evaluated methods:

- Random Forest emerged as the most accurate,
- most stable,
- and most reliable classifier.

The findings confirm that machine learning algorithms can effectively utilize maternal health indicators to predict pregnancy-related risk levels with very high predictive success.

## 11. Conclusion

This project focused on developing a machine learning-based maternal health risk prediction system using clinical and physiological indicators collected from pregnant individuals.

A complete end-to-end machine learning pipeline was successfully implemented, including:

- Exploratory Data Analysis (EDA),
- data preprocessing,
- feature scaling,
- model training,
- performance evaluation,
- cross-validation analysis,
- and feature importance interpretation.

The maternal health dataset was analyzed extensively to identify:

- statistical patterns,
- medical relationships,
- outliers,
- and clinically meaningful predictive indicators.

Several supervised machine learning algorithms were implemented and compared, including:

- Logistic Regression,
- K-Nearest Neighbors (KNN),
- Support Vector Machine (SVM),
- Decision Tree,
- and Random Forest.

The models were evaluated using multiple classification metrics such as:

- Accuracy,
- ROC-AUC,
- Precision,
- Recall,
- F1-score,
- Confusion Matrix,
- and Cross Validation scores.

The experimental findings demonstrated that all machine learning algorithms achieved highly successful maternal risk classification performance.

Among all evaluated models, Random Forest achieved the strongest overall performance with:

- Accuracy: 98.34%
- ROC-AUC: 98.63%
- Cross Validation Mean: 98.23%

The Random Forest model demonstrated:

- outstanding classification capability,
- strong generalization performance,
- minimal prediction error,
- and high robustness against overfitting.

The feature importance analysis revealed that:

- Preexisting Diabetes,
- Blood Sugar (BS),
- BMI,
- and Heart Rate

were the most influential predictors of maternal health risk.

These findings are highly consistent with established medical knowledge because:

- diabetes,
- metabolic disorders,
- obesity,
- and cardiovascular abnormalities

are major contributors to pregnancy-related complications.

The results of this study demonstrate that machine learning algorithms can effectively assist maternal health risk assessment systems by:

- improving prediction accuracy,
- supporting early diagnosis,
- and enabling proactive medical intervention.

The developed system has strong potential for:

- healthcare analytics,
- clinical decision-support systems,
- and intelligent maternal monitoring applications.



## Future Improvements

Although the developed machine learning pipeline achieved highly successful results, several future improvements may further enhance the system.

Possible future developments include:

- Hyperparameter optimization using GridSearchCV
- Deep learning approaches such as Artificial Neural Networks (ANN)
- Larger and more diverse maternal healthcare datasets
- Multi-class maternal risk prediction
- Real-time prediction systems
- Integration with hospital databases
- Web deployment using Flask or FastAPI
- Explainable AI (XAI) techniques such as SHAP analysis

Additionally, integrating:

- longitudinal patient monitoring,
- wearable sensor data,
- and real-time clinical measurements

may further improve prediction reliability and clinical applicability.

## Final Remark

In conclusion, this project successfully demonstrated that machine learning techniques can provide highly accurate and reliable maternal health risk prediction using medical and physiological indicators.

The findings highlight the growing importance of artificial intelligence and data-driven healthcare systems in improving maternal care quality, supporting medical professionals, and reducing pregnancy-related health risks.